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ACTION RESEARCH BOOKLET

By REFLACT

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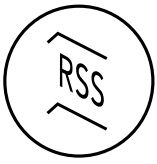


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Glossary of Acronyms

AR
Action Research
ADR
Action Design Research
BIE
Building, Intervention, Evaluation
CUAR
Critical Utopian Action Research
EAR
Ethnographic Action Research
ESG
Environmental, Social and Corporate Governance
HCI
Human-Computer Interaction
ICT4D
Information Communication Technologies for Development
IS
Information Systems
PD
Participatory Design
QWL
Quality of Working Life
SMEs
Small and Medium Entities
STS
Science and Technology Studies
VCs
Venture Capital firms



Action Research in a World of Crises and Care

Cancan Wang and Hanne Westh Nicolajsen

Who Are We and Why Did We Create the Action Research Booklet?

The Action Research Booklet has been created by the research group REFLACT at the Department of Business IT, the IT University of Copenhagen, Denmark.

REFLACT was founded in 2021, during a year when citizens, employees, organizations, and countries were tested by various crises and new orders were established. During this year, Covid-19 mutated into new variants as the vaccine arrived (World Health Organization, n.d.). Countries continued to fail to reach climate targets as new policies for climate mitigation were issued (Boehm et al., 2021; European Commission, 2021) and migration crises posed human crises and political challenges in richer countries (The United Nations Children's Fund, 2021). In the face of these crises, democratic governance across the globe continued to be eroded, with deteriorating freedom of expression and quality of elections, tightening government censorship of media, and repression of society (Papada et al., 2023). Meanwhile, more democratic forms of organization such as the cooperative – an association formed with alternative business models and shared ownership to meet common economic and social needs – regained public attention with promises

of alternative economic and political structures (Nielsen et al., 2022). What has become clear through these events is that digital technologies have become an integral part of how we organize our society, our relationships with the physical environment, and other living organisms. Not only can digital technologies potentially play an important role in tackling these crises, they can also exacerbate the status quo. Consequently, more scrutiny of how digital technologies are created, developed, and implemented in various contexts is critical for our societies and planet. As a response, we established the research group REFLACT to address the role of the digital in new ways of organizing and governing societal resources. We emphasize digitalization as a way of *collective becoming*: an approach that understands and co-creates change in a reflective and responsible way in human-technology ecosystems and infrastructures (Barad, 2007; Haraway, 1988, 2016; Haraway, 1997; Star, 1995, 2002; Star & Bowker, 2006).

In our research, we work with real-world projects and initiatives to study, intervene, and create agendas for change and innovation in human-technology ecosystems. Currently, we are working on projects that addresses the role of the digital in various areas including:

- Collective governance in decentralized ecosystems
- Democratic business and cooperative organizations
- Digital welfare and digital citizenship
- Global movements such as the green transition and open data
- Organizing and governing data commons
- Organizational resilience in global unrests such as pandemics
- Refugees and mobilization

Through our research, it has become evident that democracy is increasingly important in our discussions about digital technologies, be it in the context of welfare, business development, climate action, or refugee crises (Christensen & Ahsan, 2023; Marttila & Botero, 2021; Nielsen et al., 2022; Nino Carreras & Finken, 2022; Wang et al., 2023). We thus advocate for a care-driven, interventionist research approach to digital technologies, that supports a broader spectrum of human values in addition to generating business values. An approach that focuses on people interacting with digital technologies in different roles, such as user groups, employees, citizens, and networks of stakeholders. More specifically, we advocate for a 'reflactive' methodology that consists of both a reflective approach that utilizes participatory and field-based methods, and a proactive approach that utilizes Action Research and design methods as instruments to engage industry and society in actions. Thus, REFLACT. We categorize our research approaches under the broad umbrella of Action Research and have summarized our research traditions and experiences in a booklet that we hope will inspire a broad audience, including students, academic colleagues, and practitioners who do experiments and take part in action-oriented research in their work with humans and digital technologies.

What Is the Action Research Booklet? Who Is It For? And How Can It Be Used?

Our Action Research Booklet is a book that introduces Action Research as a diverse range of interventionist research methods on digital technologies. The booklet shares our research experiences, through which we showcase how interventionist research on digital technologies takes place in various organizational and societal contexts.

The booklet is designed as an open book where each page can be taken out and used as a card. The design is motivated by two considerations. First, the booklet is envisioned to be living and thus editable; as researchers continue to develop approaches within Action Research, emerging approaches and new experiences can be added to the booklet. This means there will be new editions of the booklet that continue to document these new explorations of Action Research. Second, the Action Research Booklet is pedagogical in nature. Different sections/cards in the booklet can be used in classrooms, for supervision, in academic discussions, or in collaborations between researchers and practitioners to explain, illustrate, and discuss different Action Research approaches.

More specifically, for university students and academic colleagues who are new to Action Research, the booklet provides a hands-on introduction to different approaches, lists key references that inspired our research, and illustrates these approaches through actual research experiences. For academic colleagues who have worked with Action Research methods, the booklet serves the additional purpose of sharing insights and challenges in using action research in different forms and new contexts.

Action Research is embedded in participation and observation practices. This means that Action Research is relevant for practice from inception to implementation. This booklet is thus also an invitation for practitioners to incorporate research in daily operations, as well as for citizens and different societal groups to affect the changes that take place in organizations, societies and the relationships with their physical environment and other living organisms, through a more 'reflactive' relationship with digital technologies.

What Is Action Research?
And How Is the Booklet Structured?

Action Research in general can be understood as a methodology that supports 'learning by doing', meaning that it contributes to the practical concerns of people in their immediate problem-situation, while satisfying the goal of scientific learning (Gilmore et al., 1986; Kemmis & McTaggart, 1988). Action Research is a research approach that emerged over time from a broad range of fields (Brydon-Miller et al., 2003). In one account, Action Research can be traced back to the 1940s, deriving from Kurt Lewin's work within organizations (Lewin, 1947). As labour-management relations gained importance, it has been widely applied in organization development, quality of working life (QWL), socio-technical systems (e.g., information systems), and organizational democracy (Carr & Kemmis, 2003; Checkland & Holwell, 1998; Nygaard & Bergo, 1975). Action Research has also gained wide popularity in the research field of education following John Dewey's philosophy that professional educators should be involved in community problem-solving (Dewey, 1938). In addition, inspired by Marxian "dialectical materialism" and the praxis orientation of Antonio Gramsci towards social change, Action Research can also be found in the circles of liberationist and feminist movements as well as international development (Borda, 2006; Frisby et al., 2009; Khene & Masiero, 2022).

Action Research has subsequently taken on other names such as Participatory Action Research (Borda, 2006) and Collaborative Inquiry (Bray, 2000), with diversities in its meanings, practices, and application contexts. Action researchers have over the years categorized Action Research in various typologies that help to orient the similarities and differences between these approaches

(e.g., Davison et al., 2021; Kemmis, 1988; McNiff, 2013). In our work on digital technologies and how they affect changes in our society and organizations, the Action Research approaches we encountered generally span from a functionalist to a critical approach. The utilitarian, functionalist position investigates how digital technologies can be used to enhance and optimize existing processes. This perspective is more common in, for instance, Information Systems studies that deal with the use and development of IT in organizations, without necessarily being concerned with power structures within organizations (Baskerville & Wood-Harper, 1996). The critical position raises awareness of the multiple interests different actors have in societal or organizational contexts, the power dynamics between these different groups of actors, and the representation of certain groups in the design, development, and implementation of digital technologies (Borda, 2006; Frisby et al., 2009; Khene & Masiero, 2022). This also means that certain Action Research methods can be used in various ways. For instance, the method of Future Workshops may be used in a functionalist or a critical Action Research approach (Jungk & Müllert, 1987; Nicolajsen et al., 2016).

However, in both approaches, interdisciplinary perspectives have proven to be relevant in understanding the interplay between digital technologies, organizations, and society. To do so, we begin in this booklet by introducing different disciplinary traditions of Action Research. Starting from Critical Utopian Action Research, Vasiliki Baka takes the reader on a historical journey to understand how Action Research has been developed with a committed democratic impulse in Nordic countries such as Denmark, and in the Global South since the 1980s. This is followed by an introduction to two different Action Research approaches in Information System

studies, with an organizational focus on the development and use of digital technologies. While Action Research was initially brought in to balance between practical relevance and theoretical rigor in Information Systems studies, a later development – Action Design Research – focuses more on generating prescriptive design knowledge through building and evaluating IT artefacts in an organizational setting (Sein et al., 2011). Drawing on comparisons with Action Research in Information Systems studies, Lene Nielsen and Laura Juncker-Jensen introduce Action Research in two adjacent fields to Information Systems – Human-Computer Interaction (HCI) and Information and Communication Technologies for Development (ICT4D). In both Nielsen and Juncker-Jensen's accounts, despite a primary focus on intervention, Action Research was less pronounced in HCI and ICT4D in comparison to Information Systems due to different trajectories of field development. Nielsen introduces a taxonomy of Action Research to better understand the differences between a diverse range of Action Research approaches in HCI and Information Systems. Building on an introduction to the development of the Action Research methods used in ICT4D, Juncker-Jensen provides a critical perspective on why and how Action Research is needed for ICT4D.

Sanna Marttila and Sisse Finken compare Action Research to another participatory research methodology – Participatory Design. In this chapter, they introduce the field of Participatory Design as one that "explores possibilities and conditions to enable user participation in the design of technologies and information systems" (Marttila & Finken, 2024: ch. 2.6), and account for similarities and differences between Action Research and Participatory Design. They conclude that while both approaches share participatory frameworks, principles, and values, Participatory Design emphasizes

the integration of collaborative design elements into the research process. Drawing on such comparison, Sisse Finken further discusses the performativity of scientific methodologies, and within this, the art of paying attention to positions and power when engaged in designing ecologies for humans, more-than-humans, and technologies.

This is followed by two pieces that provide an account of what cuts across these traditions, one articulating shared methods such as Future Workshops, and one addressing the role of the action researcher. Turning back to the democratic focus on digital technologies, Hanne Westh Nicolajsen introduces Future Workshops as a method in Action Research that is specifically used to involve multiple groups of stakeholders in the search for solutions to societal and organizational challenges. Section 2 ends with a general reflection by Hanne Westh Nicolajsen and Vasiliki Baka on the role action researchers play in various research contexts, providing a compass for action researchers to reflect on their identities in the field and to position themselves accordingly.

Today, Action Research continues to be developed in various research and empirical contexts, where the repertoire of methods and their applications become more diverse. In the second half of the book, we therefore also include specific Action Research experiences where existing methods are being deployed in unprecedented contexts and take on new forms, and where new methods are also emerging following different application scenarios. More specifically, Hanne Westh Nicolajsen, Sisse Finken and Lars Rune Christensen introduce their research experiences regarding the development of telecare in a public-private collaboration between a municipality and a technology provider. Through the case study, they detail their

experiences of organizing a future workshop and reflect on their roles as experts in the field. This is followed by Lene Nielsen, Hanne Westh Nicolajsen and Vasiliki Baka's account of their Action Research experience regarding organizational challenges and solutions for democratically owned businesses. In this chapter, they share their experiences of organizing a two-day workshop and inviting practitioners to share their organizational experiences with a focus on collective learning. The section ends with an Action Research experience where researchers and a local business association in Copenhagen experimented with a governance concept – 'data commons' – intended to motivate small businesses to voluntarily work towards the sustainability transition. In this chapter, Yvonne Dittrich, Qiuyu Jiang, Cancan Wang, Laura Juncker-Jensen and Carsten Østerlund detail their experiences of using a governance concept as a method of intervention. They share key learnings from their research experiences and discuss the importance of concepts and artefacts as probing devices for understanding and intervention, as well as mediating conditions such as boundary spanners, boundary objects and organizational practices.

As researchers explore new contexts for Action Research, we end the first edition of the book with an open discussion through Vivian Wei Guo's PhD project on climate data activism in China, where she envisions how Action Research can help to rethink the way researchers participate in the field, as well as the value of our research.

Have a good read!

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2.1 Critical Utopian Action Research

Vasiliki Baka

Critical Utopian Action Research (CUAR) is a subdivision of Action Research that was developed in the early 80s, initially in Denmark, but over the years has also been established in other Scandinavian countries and countries of the Global South. In fact, the initial project that adopted CUAR can be considered the project 'Industry and Happiness', set within the Danish fishing industry and carried out by a small group of researchers together with female workers (Gunnarsson et al., 2016).

The theoretical underpinnings of CUAR can be traced back to the 'old Frankfurt School', but it is also highly inspired by Critical Theory and the work of Adorno, Horkheimer and Bloch, as well as critical social theorists like Oskar Negt and Regina Becker-Schmidt (see Bladt & Nielsen, 2013). As the term implies, both 'critical' and 'utopian' perspectives are emphasized theoretically and methodologically. On the one hand, CUAR intends to create "critical awareness about the necessity of change" (Coghlan & Brydon-Miller, 2014: p. 2; Rom & Green, 2023) and on the other hand, utopianism is promoted through workshops during which participants are encouraged to create their ideal futures and visions. Wright (2013: p. 3) nicely presents how utopia is to be perceived:

> "[U]topia is thus both a nowhere place and a good place. It is the fantasy of a perfect world that fully embodies our moral ideals [...] The idea of real utopias embraces this tension between dreams and practice: utopia implies developing visions of alternatives to dominant institutions" (in Rom and Green, 2023).

It is this exact desire to create an ideal life that mobilizes participants: *How do we really want to live, how should we live?* (Egmose, et al, 2020). Such questions need to be answered if we are to meaningfully contribute to processes of democratization. Democracy as both a means and an end is closely related to the perception of a utopian future, whereby citizens have the necessary space for personal and social development. CUAR acts thus as a systematic methodological framework that aims at increasing bottom-up engagement by actively involving citizens/research participants in decision-making and agenda setting.

This systematization is made manifest through three stages, namely critique, aspiration, and action. As expected, given that CUAR is part of AR, it is still committed to the "initial democratic impulses of Lewin and Freire", yet highly enriched with elements from the work of Robert Jungk and especially his Future Creating Workshops (Nielsen & Nielsen, 2016). Future Creating Workshops are organized so that, first, the important phase of critiquing existing practices is initiated, then the utopian phase of resolving problems follows, with the last step being the realization phase of actually developing implementation plans (Coghlan & Brydon-Miller, 2014: p. 4). Birger Steen Nielsen and Kurt Aagaard Nielsen, who were among those who developed CUAR as a methodological practice, utilized their experience in facilitating Future Creating Workshops combined

with their interest in Critical Theory to open the way for researchers with the same mandate, namely, to empower people to realize ideal futures. Over the last 40 years several cases have been reported, ranging from day care centres (Ahrenkiel, 2016) to local communities and workplace development (Rom & Green, 2023).

All in all, CUAR differs from other forms of Action Research in that it emphasizes and develops even more powerfully "the democratic and social commitments of Action Research as fundamental to a prosperous and peaceful societal development" (Nielsen & Nielsen, 2016).

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2.2 Action Research in Information Systems

Cancan Wang

Information Systems (IS) is a field that deals with the development and use of IT artefacts in organizations. With its iterative approach to change, Action Research in IS primarily focuses on "helping practitioners" implement organizational change and/or development of IT systems and software, while simultaneously studying the process (Baskerville & Myers, 2004, p. 330; Hirschheim et al., 2012). Overall, Action Research in IS is strongly oriented towards collaboration and change involving both researchers and practitioners. Making use of an iterative process means that emphasis is on learning on the part of the researchers and practitioners who are situated in a particular research context.

Action Research was introduced to IS largely due to the field's long-term struggle with balancing practical relevance and research rigor. Advocates argue that Action Research is an important methodology for IS researchers to consider, given its potential to expand scientific knowledge whilst assisting in solving practitioners' problems (Baskerville & Wood-Harper, 1998; Mathiassen, 2002; Sein et al., 2011).

Action Research in IS draws on a wide range of previous method development in different research traditions and contexts such as

organizational development practice (e.g., Carr & Kemmis, 2003), social psychology study (e.g., Lewin, 1947), participatory design (Mumford & Weir, 1979), socio-technical system research (e.g., Checkland & Holwell, 1998; Wood-Harper, 1985) and the Nordic trade union research tradition (e.g., Nygaard & Bergo, 1975). In general, Action Research in IS initially shares a philosophical root with pragmatism that concentrates on "asking the right questions and getting empirical answers to those questions" (Baskerville & Myers, 2004, p. 331). According to Richard Baskerville and Michael D. Myers (2004), there are four key premises that stem from pragmatism, including:

- "Consequences define human concepts": this means action researchers need to explicate theory before action is taken
- "Practical outcome embodies truths": Action Research emphasizes the importance of action as embodiment of truths. Theory must be validated by its practical outcome
- "Logic of controlled inquiry": Practical actions and rational operations must be interpreted for learning to take place
- "Human action and interpretation are contextualized socially": Action Research needs to be conducted in a collaborative team through reasoning, action formulation and action-taking

Today, drawing on different research disciplines and contexts, Action Research in IS has also developed more nuanced characteristics in practice. For instance, some researchers advocate for Action Research as a "clinical method" (Baskerville & Myers, 2004, p. 330), or "a theory-driven research lens is used to intervene in a problematic situation" (Davison et al., 2021 , p. 851). This clinical method is often characterized by a clearly distinguished research stage, where a diagnosis is made before intervention and reflection. In chapter 2.4,

we provide a comparison between the “clinical method” and the Action Research approach in human-computer interaction.

According to Mathiassen (2002), some researchers took their point of departure in the different types of knowledge created in collaborative research processes between researchers and practitioners in systems development, and advocate for a pluralist research methodology. This line of thought emphasizes that researchers should not play the role of consultants in Action Research projects (though the authors do not clarify what a consultant role may or may not consist of). Rather, they argue there are three research goals and corresponding research practices, which researchers can engage with during collaborative system development projects.

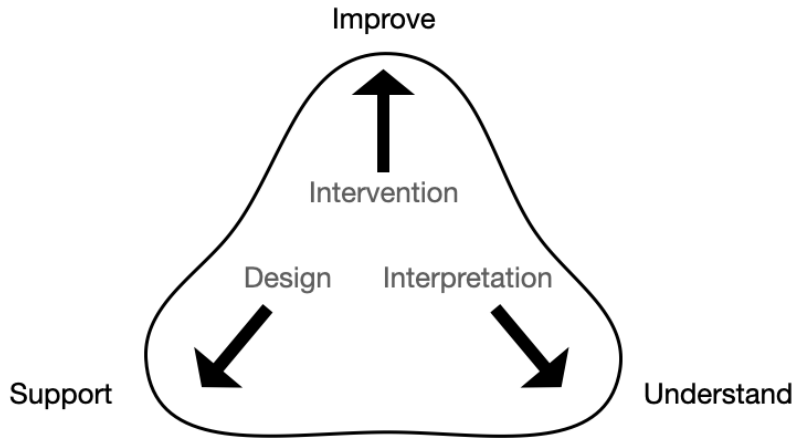


Figure 1: Research Goals and Activities (Mathiassen, 2002, p. 327, adapted from Vidgen & Braa, 1997)

First, to *understand* systems development, researchers collect data about professional practice and interpret these using different concepts and frameworks, resulting in insights into systems

development practice and a theoretical toolkit that can be used to reflect upon systems development practice.

Second, to build new knowledge that can *support* professional practice, researchers design normative propositions or artefacts, e.g. guidelines, standards, methods, techniques, or tools, to plan, guide, or improve practice.

Third, to learn how to *improve* practice, researchers engage in different forms of social and technical intervention to observe changing practices, both to uncover existing assumptions in professional practice and generate insights into barriers to and enablers of systems development improvement efforts.

In his article on collaborative practice research, Lars Mathiassen (2002) argues that these different research activities support each other but can also be pursued separately. Depending on the goal of the study, researchers can combine Action Research with qualitatively different research methodologies (e.g., experiments or case studies).

Others again combine Action Research with intervention-oriented subfields in IS such as Design Science (Sein et al., 2011). Such combination, often referred to as Action Design Research, emphasizes that an IT artefact is an ensemble of material and organizational features, which means that identifying new technological solutions is simultaneously an organizational intervention. Action Design researchers such as Maung K. Sein, Ola Henfridsson, Sandeep Purao, Matti Rossi and Rikard Lindgren (2011) argue that reflection and learning do not take place after an intervention; rather, they occur continuously during problem framing and shape the emerging ensemble artefact.

Finally, they highlight the importance of abstracting knowledge generated in a particular problem-instance into general design principles for tackling a class of field. In chapter 2.3, we introduce Action Design Research in more detail.

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2.3 Action Design Research in Information Systems

Cancan Wang

In this chapter, I will introduce another stream of Action Research in Information Systems - Action Design Research (ADR) (Sein et al., 2011). Following an initial definition of ADR and its foci, I will give an account of how ADR has become a prominent strand of Action Research as a response to the traditional Design Science research tradition. This is followed by a presentation of the archetypical stages and principles of ADR.

ADR is "a research method that generates prescriptive design knowledge through building and evaluating ensemble IT artefacts in an organizational setting" (Sein et al., 2011, p. 40). ADR specifically deals with two challenges: 1) addressing a *problem situation* in a specific *organizational context* through intervention and evaluation, and 2) constructing and evaluating an *IT artefact* that addresses the class of problems typified by the encountered situation. To address these two challenges, ADR places emphasis not only on the theoretical precursors and the intent of individual researchers, but also on the influence of users and ongoing use *in context*.

ADR arises against the traditional Design Science research tradition that focuses on generating design knowledge about *innovative IT artefacts* for practitioners. In the Design Science research tradition,

researchers seek to develop prescriptive design knowledge through a "build and *then* evaluate" cycle, moving linearly from identifying a class of problems, to goals, to conceptualization, to designing, to building, to evaluating innovative IT artefacts. As such, Design Science research promises two results: a specific IT artefact, and the knowledge about creating other instances of IT artefacts that address the same class of problems. In their seminal piece on ADR, researchers Maung K. Sein, Ola Henfridsson, Sandeep Purao, Matti Rossi and Rikard Lindgren criticize the Design Science research tradition from two aspects: 1) the separation between building and evaluating, and 2) the priority of IT artefact invention over organizational intervention (ibid.). As a response, they argue for the expansion of Design Science research methods with the ADR approach, which explicitly accounts for the role of organizational context in the design of innovative IT artefacts. Rather than viewing IT artefacts as material features, ADR researchers view *IT artefacts as an ensemble of both material and organizational features*. IT artefacts are therefore socially recognized as bundles of hardware and/or software (Orlikowski & Iacono, 2001). As such, ADR researchers argue that the ensemble artefact should reflect both the intended and unintended organizational consequences. This means that the building and use of an IT artefact cannot be separated. And instead of solely being guided by researchers, users should also play an important role in designing and building an IT artefact. Drawing inspiration from Action Research, ADR is thus envisioned as an iterative process that takes consideration of organizational contexts when building IT artefacts, links theory with practice, and combines the viewpoints of researchers and practitioners. More specifically, ADR follows four stages and seven principles as shown in Figure 1 on the next page.

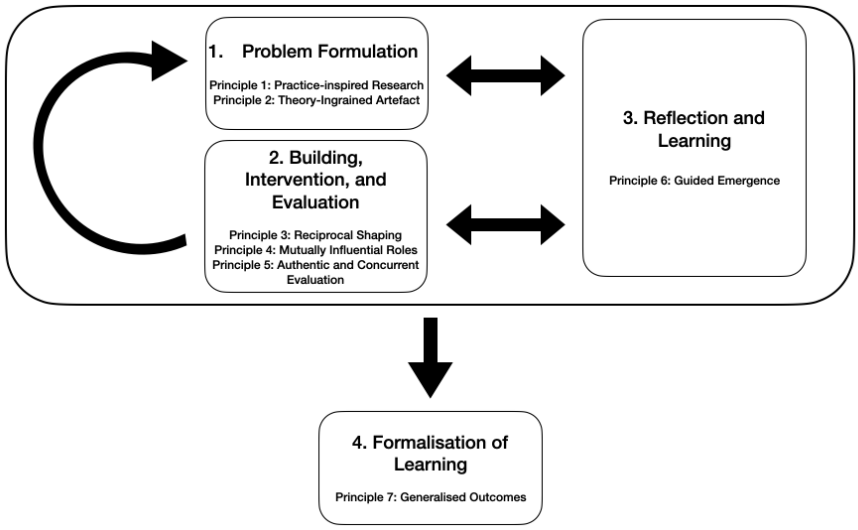


Figure 1: ADR Method: Stages and Principles (Sein et al., 2011, p. 41)

ADR starts with Stage 1 - Problem Formulation, where a problem is perceived in practice and/or anticipated by researchers. During this stage, researchers and practitioners determine the initial scope, decide the roles and scope for practitioner participation, and formulate the initial research question.

Two key principles at this stage are *practice-based research* and *theory-ingrained artefact*. This means, on the one hand, researchers need to view field problems at the intersection of technological and organizational domains as knowledge creation opportunities. But this does not mean that researchers take on the role of a consultant or a software engineer. On the other hand, researchers actively inscribe theoretical elements in the ensemble artefact, manifesting the theories "in a socially recognizable form" (Orlikowski & Iacono, 2001, p. 121). Theories at this stage are used to structure the problem, to identify possible solutions, or to guide design. And they often only

result in the early design of the ensemble artefacts, which are then further shaped by organizational use and subsequent design cycles in Stage 2.

For example, the primary tasks during Stage 1 could include (Sein et al., 2011, p. 41):

- Identify and conceptualize the research opportunity
- Formulate initial research questions
- Cast the problem as an instance of a class of problems
- Identify contributing theoretical bases and prior technology advances
- Secure long-term organizational commitment
- Set up roles and responsibilities

Drawing on the problem framing and theoretical premises of the IT artefact, Stage 2 interweaves *Building* of the IT artefact, *Intervention* in the organization, and *Evaluation* (BIE). Stage 2 is carried out as an iterative process in a target group, where both the *problem* and the *artefact* are continually evaluated. Equally importantly, this stage also clarifies what the locus of innovation is in the ADR projects, which is often a continuum between artefact design and organizational intervention.

To give some examples, when the ADR team primarily focuses on creating an innovative technological design, there can be at least two cycles – first, lightweight interventions in a limited organizational context with early designs to test the participants' assumptions, expectations, and knowledge. As such, initial cycles help to build organizational commitment and guide the design of the ensemble artefact. They are followed by comprehensive interventions in a wider organizational context with a more mature artefact, which allows for

large-scale evaluation in the use setting and ongoing refinement of the ensemble artefact.

When the ADR team primarily focuses on creating innovative organizational intervention, they deploy the artefact in organizations in its early iterations. They improve the artefact design by challenging organizational participants' existing ideas and assumptions about how to use a specific artefact. These iterations end with assessments of the artefact and the design principles it represented.

Stage 2 emphasizes the inseparable influence of the two domains, IT artefact and organizational context, on each other, the mutual learning between researchers and practitioners, and ongoing evaluation during (instead of after) building. Overall, these iteration cycles result in the realized design of the artefact. The primary tasks in stage 2 can include:

- Discover initial knowledge-creation target
- Select or customize BIE form
- Execute BIE cycles
- Asses need for additional cycles, repeat

During Stages 1 and 2, constant reflection and learning (Stage 3) take place. Here, Sein et al. (ibid.) particularly refer to the importance of theoretical reflection, learning and contribution in addition to problem solving in ADR. Knowledge contribution is made possible during conscious theory-practice dialogue during problem framing, when choosing theories, and in shaping the emerging ensemble artefact.

'Guided emergence' is thus an important principle during this stage.

It emphasizes that knowledge creation takes place between *design* as "external, intentional intervention", and *emergence* that seems to imply "organic evolution" (ibid., 2011, p. 44). More specifically, guided emergence highlights that the ensemble artefact is shaped by the preliminary design co-created by the researchers, the organizational use, perspectives, and participants, as well as outcomes of authentic, concurrent evaluation.

By the end of an ADR project (Stage 4), researchers are required to have generalized the knowledge created in a particular problem-instance into solution concepts for a class of field problems. At the last stage, the researchers outline the accomplishments realized in the IT artefact and describe the organizational outcomes in order to formalize the learning, which can act as a design principle for future solution design for the same class of field problems. In generalizing such outcomes, Sein et al. suggest conceptualization can take place at three levels: "1) generalization of the problem instance, 2) generalization of the solution instance, and 3) deviation of design principles from the design research outcomes" (ibid., p.44).

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2.4 Action Research in Human-Computer Interaction and Information Systems

Lene Nielsen

Action Research has many faces, but common to them are the emancipatory and learning aspects of the interventions. In the following, we will present the three different perspectives that all create changes within the domain of digital technology.

Action Research has never played a dominant role within the field of Human-Computer Interaction (HCI), one reason might be that the field has stronger ties to Participatory Design (PD) than to Action Research. Both PD and Action Research stem from the notion that change should be designed and implemented democratically and inclusively (Foth & Axup, 2006). However, the scope of PD is typically more limited to the design of solutions, whereas the scope of Action Research includes the notion of learning through action (see chapter 2.6).

Contrary to this, Information Systems (IS) has a long tradition for Action Research and in the following we will look at both HCI and IS and the way they propose to do Action Research projects.

According to Grundy, Action Research projects can be understood according to a taxonomy that divides the projects into three different

domains with distinct goals: technical, practical-deliberative/critical-emancipatory, and emancipatory (Kemmis, 1988).

Technical or scientific-technical is traditionally most relevant to the natural and computational sciences, in which truth and reality are generally thought to be knowable and measurable, and knowledge produced may be predictive and generalizable.

The facilitator engages with collaborators to test an intervention based on a predefined scientific theory. This intervention is designed to create some change in the setting, which can include new practices and approaches, different power structures or group dynamics, altered patterns of action, or simply the incorporation of a new piece of technology into daily practice.

Practical-deliberative/critical-emancipatory is more familiar to the humanities or a critical theory research team. It focuses more on the unknowable and social realities, with research problems that are constantly evolving and defined in the situation by a variety of stakeholders with dynamic and mixed values. Practical Action Research is largely about understanding local practices and solving locally identified problems.

Emancipatory Action Research promotes a kind of consciousness-raising and criticality that seeks to empower partners to identify and revolt against problems they may not have identified initially on their own.

Focusing on the Scientific-Technical

Examples of authors who work specifically within the technical

domain are Hayes (2014) and Baskerville and Wood-Harper (1996). Hayes (2014) promotes a process with six steps:

1. Establishing a Relationship with a Community Partner
2. Research Questions and Problem Statements
3. Action and Intervention
4. Evaluation
5. Disseminating Knowledge and Documenting Progress
6. Leaving the Site

Similarly, Baskerville and Wood-Harper advocate for a process with five distinct characteristics:

1. A varied social environment
2. Hypothesis formation about observations
3. The researcher's intervention
4. Participant observations
5. The study of change in the social environment

As can be seen from the two processes, Hayes (2014) is more concerned with the actual process of interacting with the community partners, while Baskerville and Wood-Harper (1996) focus on the change in the environment. This perspective is furthermore present in their division of Action Research into two stages: a diagnostic stage, where the situation is analysed, and a therapeutic stage, where the change experiments take place, and the effects are studied.

Davison et al. (2021) list different methods that can be integrated into what they term canonical Action Research, which Action Research can learn from. From the integration 11 new forms of Action Research arise:
1. Action Design Research – focus on the inclusion of IT artefacts

2. Action Learning - focus on conversations with clients
3. Action Science - focus on the development of both skills and confidence among members of organizational teams
4. Clinical Fieldwork - where researchers interact with paying clients, behaving in a professional manner is a key to success for researchers.
5. Dialogical Action Research - focuses on the immersion of the researcher in the world of the clients
6. Grounded Action Research - focuses on inductive theory development from data collected during the project
7. Networks of Action - researchers should be aware of an implicit responsibility to ensure that their interventions do not merely satisfy short-term research needs and client requirements
8. Participatory Action Research - the focus of this method is on collaboration and reflection
9. Participant Observation - participatory observation requires the researcher to be immersed in the problem situation
10. Statistical Action Research - the focus is on the development of a formal theoretical model that can be tested with statistical techniques to ascertain whether an intervention has achieved the desired effect
11. Soft Systems Methodology - the building of conceptual models is a useful technique

Many of these do not include all the five or six phases introduced by Hayes and Baskerville & Wood-Harper but have a specific focus.

Action Research aims to improve organizational problem situations for all relevant stakeholders and pays attention to the interests of all stakeholders involved in the project. However, often the perspective of the researcher is not explicitly discussed, as was found in a literature study by Clarke & Davison (2020).

This can be seen in the suggestion to blend Action Research with other methods such as case studies and surveys (Davison et al., 2021). The argument is that if it can be demonstrated that Action Research complements case studies and surveys in ways that lead to richer insights and more significant impacts, then it will be more appreciated by reviewers and editors. This is mainly for the benefit of the researchers and not all stakeholders.



Image 1: A student group prototypes a future digital service design process.

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2.5 Action Research in Information and Communication Technologies for Development

Laura Juncker-Jensen

The academic field of Information and Communication Technologies for Development, commonly referred to as ICT4D, emerged from within the field of Information Systems in the 1980s and early 1990s (Heeks, 2008; Walsham, 2017). It explores how information systems and technology can be applied to various contexts in the Global South (Ramadani et al., 2018), with the inherent purpose, as indicated by the name 'for development,' of positively transforming socio-economic conditions in Global South countries (Avgerou, 2010). The premise of both the research and practice in the field is that ICTs can be leveraged to achieve certain outcomes, although the ways in which this happens are subjects of dispute across different schools within the field (Heeks, 2008; Maail, 2011). ICT4D has since its inception expanded to span multiple disciplines such as development studies, computer science, and subfields including Human-Computer Interaction, geography, anthropology, and community informatics (Sterling & Rangaswamy, 2010; Walsham, 2017).

Among the methodological choices an ICT4D researcher is faced with, a number of circumstances arguably pose challenges and demands that are distinct from practice-oriented Information Systems (or related) research in Global North countries. This among

other things includes the close engagement with contexts of social and economic vulnerability (Khene & Masiero, 2022), concern with the effects on a given society of a project usually instigated from an outside context, measuring success in terms of complex social indicators, rather than, say, organizational performance, and by likely implicating power imbalances between participants in the ICT project - i.e., the researchers on the one hand and the users on the other hand (Sterling & Rangaswamy, 2010). Although such circumstances are not always factored into the planning and roll-out of ICT4D projects (which have been criticized for often being Northern- and techno-centric) (Heeks, 2008; Khene & Masiero, 2022), they require special skills, knowledge, and organization to achieve a positive influence in the highly complex context of lower-resource environments. Or at the least, to avoid producing adverse socio-economic outcomes for the intended beneficiaries (Bon, 2019).

Despite a strong emphasis on intervention in ICT4D, there is not a widespread tradition of using Action Research methodology for designing ICT4D research and practice (ibid.). A possible explanation for this might be that ICT4D research is already closely related to practice, while these are typically treated as more separate components in other academic disciplines (Sterling & Rangaswamy, 2010). The implicitness of practice in ICT4D research might thus, to some eyes, obviate the need for explicitly applying, interrogating, and developing ICT4D-specific Action Research methodologies.

Another contributing factor might be that the field suffers from a lack of theorization, not only regarding methodology, but as a general characteristic, due to this orientation towards practice (Abbott & Wagg, 2023; Bon & Akkermans, 2019). As Heeks (2007, p. 1) put it:

> "...The explosion of work on ICTs for development (ICT4D) has (unconsciously) followed Marx's dictum: "The philosophers have only interpreted the world differently; the point is, to change it." There has been a bias to action, not a bias to knowledge. We are changing the world without interpreting or understanding it. Most of the ICT4D research being produced is therefore descriptive not analytical."

Some Uses of Action Research in ICT4D

Nonetheless, a steady stream of Action Research studies has been published in ICT4D, including among others, long-standing work from the Oslo group in Health Information Systems in several countries, from South Africa in various domains, and from Amsterdam -based scholars focusing on rural development in the Sahel (Bon & Akkermans, 2019). Bon & Akkermans (ibid.) summarize the use of Participatory Action Research in such works in a set of principles that is adhered to by both Information Systems and ICT4D research:

Guiding principles of Participatory Action Research in Information Systems and ICT4D (ibid.)

1. Principle of Critical Investigation of Concrete Situations (field, professional practice)
2. Principle of Value: Developing/Taking a Value Position (democracy, emancipation, autonomy, social and economic betterment)
3. Principle of Stakeholder Collaboration (involving Co-Investigation, Co-Design, Co-Creation, whereby goals and interests as seen by stakeholders themselves are central)
4. Principle of Dialogue (between multiple actors and stakeholders (to be) involved)

5. Principle of Action: Discovery and subsequent Realizing of Change for the Better
6. Principle of Reflection and Continuous Learning in Action

More interestingly, a number of authors have in recent years been contributing to the theoretical and practical advancement of various aspects of Action Research methodologies designated for ICT4D initiatives. For instance, Anne Bon (2019) addresses the conundrum of increased demands for *contextual embedding* in environments that are characterized by resource scarcity. While the importance of situating technical work in its broader organizational contexts has for some time been a central component in the work of information system developers and engineers in ICT projects in more affluent parts of the world, what this means for working in a development context is still in need of its own strand of writing in the ICT4D literature (ibid.). Bon draws on complexity theory to address non-linearity and unpredictability in ICT4D initiatives, in stark contrast to the assumptions of predominant models in ICT4D (and development) interventions, such as Theory of Change and Logical Framework Approach. The latter are used to describe a list of steps leading from problem to development goal following a *linear* perspective on activity and effect. With a nod to the shift in Global North ICT sectors from using the linear Waterfall model of the 60s and 70s to more agile models allowing for flexibility in project progress today, Bon has developed a flexible methodology for ICT4D based on various sources from within Action Research. This methodology emphasizes the non-linearity of socio-economic development and entails *context analysis, needs assessment*, and *use case and requirements analysis* that are attuned to the specific demands of a lower-resource-context, and that cover the complete lifecycle of information systems engineering (ibid.). Here, learning

from the process to improve it continuously through iterations and 'learning loops' is imperative to avoid inappropriate (albeit historically much-used) deterministic and mechanistic planning modes for ICT4D initiatives (ibid.).

Another methodology is Ethnographic Action Research (EAR), which is a form of Participatory Action Research specifically designed for ICT4D initiatives with an emphasis on media and communication technologies (Tacchi, 2015). It integrates ethnographic approaches for a more immersive and holistic understanding of local contexts through long-term engagement, and uses continuous planning, action, and reflection as a way of navigating ever-changing projects. Ethnographic Action Research and its foundational concept of Communicative Ecologies (see more in ibid.) are a counter-reaction to two organizing principles that tend to dominate ICT4D initiatives; first the techno-determinist and modernizing school of thought, where technology is elevated as an intrinsic tool for development, and second the adherence in the field to *results-based management*, which values measurement and impact. EAR challenges predominant assumptions about ICTs and development by embedding the understanding of both within surrounding social structures. Concretely, the methodology sheds light on how individuals and communities *communicate* around certain themes or tasks, to fully grasp which communication channels are used and why, and how this constitutes opportunities and barriers to introducing any new social and technical connections in the ICT4D initiative, making these more likely to adapt to existing local structures and thus to succeed (ibid.).

Over and above theory-driven design of research endeavours, in ICT4D, Action Research is also considered, and widely used as, a useful means of creating knowledge through *post hoc theorization,*

although this mostly is in connection with large-scale infrastructure design and development projects (Abbott & Wagg, 2023). Post hoc theorization entails a sensemaking process of applying, and engaging with, theoretical or conceptual perspectives *after* the research has been conducted, in acknowledgement of its own unplanned nature. The way it is often deployed is firstly by immersing in the local context, and then drawing on a few theoretical concepts to further theorize using insights from the studies. This helps shed light on how and why socio-technical changes may take place in a certain context (ibid.).

A Note on Participation

Across the general field of ICT4D, failure, in the sense of not achieving the intended outcomes, continues to define ICT4D initiatives (Harris, 2016; Heeks, 2008). Among explanatory factors, some highlight the linear, modernist, and top-down approaches of some ICT4D interventions, a discrepancy between deployed technologies and the actual local needs, and notably the failure to pursue or obtain sincere inclusion of users and beneficiaries of the ICT in the conception and design of the project (Bon, 2019; Tacchi, 2015).

Participatory methods are commonplace in ICT4D and participation is generally acknowledged as crucial to the long-term sustainability of ICT4D initiatives (Gomez, 2013; Maail, 2011). Although the involvement and active participation of all groups of intended users in a particular community are considered basic prerequisites for ensuring that they can all benefit (Maail, 2011), some have cautioned that the often superficial uses of the term participation only amount to a mere buzzword that is incorporated mainly to satisfy donor requirements (ibid.). Therefore, a (somewhat sporadic and not

exhaustive) list of reflections follows, for the consideration of further developing the concept of participation within Action Research in ICT4D.

· **Thinking of participation as a decolonial effort.** The practice of taking decolonial approaches to ICT4D is fairly recent (Khene & Masiero, 2022). Not thoroughly addressing Western biases in ICT4D research puts it at the risk of losing out on contextual nuance and of failing to give voice to the intended beneficiaries. Approaching participation as a way of incorporating epistemologies and value systems that are indigenous to the intended beneficiaries, can serve to deconstruct Western-based concepts, and reconstruct indigenous ones in order to lay the foundations for Action Research-based planning of ICT4D initiatives (ibid.)

· **Thinking of participation as a balancing of power.** In a similar vein, the individual meetings and relations between researchers and intended beneficiaries in ICT4D initiatives are likely to imply power imbalances. When designing an engineering or research project with the primary purpose of benefiting a certain community, methodologies aimed at creating level power structures between project managers and community members (e.g. through thorough informed consent - Sterling & Rangaswamy, 2010) can have the advantage of putting the best interest of participants first, regardless of performance on other success indicators (timeliness, budget, product performance etc). Decolonial methodologies also offer inspiration here (Thambinathan & Kinsella, 2021; Smith, 2012).

· **Thinking of participation as embedding user knowledge in the ICT itself.** Viewing participation as a question of merely including users in the process of developing or implementing ICTs, takes focus away

from thinking of participation as part of the ICT itself. Following Bijker's (2006) conceptualization of technology as not only artefacts or activities, but also as institutionalized forms of knowledge, allows the research to go beyond approaching the ICT as a full or partial fix to a given socio-economic problem, to considering what knowledges it serves to stabilize and whose knowledges they are.

· **Thinking of the enactment of participation as unpredictable.** Referring to participation without conceptualizing its form in more detail can potentially veil a range of concrete activities and behaviours when deployed in practice. Conceptual vagueness of the word can therefore render it amorphous in terms of how users "participate when participation is what is expected from them", as also seen in 'patient participation' in e-health (Nielsen & Langstrup, 2018, p. 1). Here, the enactment of participatory practices can take the form of various beneficiary tactics for engaging creatively with the invitation to participate, which might very well be true for participation scenarios in ICT4D as well. Rather than aiming for a certain type of engagement from users and beneficiaries, having an awareness of the social complexity of participation itself might attune the researcher to unpredictability in the ICT4D initiative as part of her methodological approach.

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2.6 Action Research in Participatory Design

Sanna Marttila and Sisse Finken

In this chapter, we explore two distinct participatory research methodologies: Participatory Design and Action Research. While both approaches converge on shared participatory frameworks, principles, and values, Participatory Design integrates collaborative design elements into the research process.

The field of Participatory Design (PD) explores possibilities and conditions to enable user participation in the design of technologies and information systems. Since the 1970s, PD has aimed at involving future users in the early design phase of computer-based systems with a democratic and political objective. At that time computers and technology systems were slowly becoming an integral part of working life. The early PD projects in Scandinavia aimed to increase democracy in the industrial workplace by inviting employees – not only the management – to be directly involved in the design processes. With this new approach to systems development, new development practices, tools and methods were also introduced (for overviews of the history and development of PD see e.g. Ehn & Kyng, 1991, Robertson & Simonsen, 2012).

Since the early decades of PD, the field has matured and moved beyond the work context, as people's everyday lives are everywhere

shaped by information technologies and mediated through digital materials. In tandem, the professional design practice has moved to the offices, homes, and other domains of our private and public life. Consequently, this move had implications for the design practice, which was increasingly "seen as continually on-going, and intricately interwoven with use" (Henderson & Kyng, 1992, Dittrich et al., 2002). Using Action Research strengthened questions regarding accountability in PD, and it strengthened how PD researchers' work should contribute to local needs, rather than primarily to the research community (Suchman, 2002).

> "Participatory design has always given primacy to human action and people's rights to participate in the shaping of the worlds in which they act" (Robertson & Simonsen, 2012, p. 4).

Since the beginning, Action Research has been an incremental part of PD practice and projects, especially "in terms of attempting to change situations, not simply study them" (Robertson & Simonsen, 2012 p. 36, 117). As we have learnt in this booklet, Action Research is a practical research methodology that aims to produce new knowledge and develop theory through reflective practice by combining scientific and action inquiry (cf. Greenwood & Levin, 1998, Reason & Bradbury, 2001). At its core, like PD, Action Research aims to challenge the current context of investigation (e.g. system, practice, environment), and, through collaborative design, make changes.

Action Research and PD seem to share almost identical objectives, values, and approaches, but some PD scholars argue that even though they share a similar interest in participation, they are "quite different in their intent and purpose", and PD projects could benefit from and build more upon Action Research methodologies (Foth

& Axup, 2006, p. 93). For example, in traditional Action Research, research results are commonly shared among participants, while often in Action Research-influenced PD, the research results are actions and experiments, more than reports or research papers that could be easily distributed. These knowledge outputs are only useful if they are geared for action (Reason & Bradbury, 2001). Therefore, it is essential to document the design research process rigorously from the design challenge to the solution, as it enables others to assess the findings and decisions made (Swann, 2002).

Applying Action Research in design research and practice is often advocated for within PD, and its influence for PD has been widely acknowledged within the field (ibid., Spinuzzi, 2005). In PD and Action Research alike, both the (design) process and artefacts are in the process of being shaped. According to Greenbaum & Loi (2012), in Action Research the point of departure is the needs and wishes of the participants, whereas researchers support these needs and wishes with a view to meeting the participants' and researchers' collectively shaped objectives. In turn, the PD approach brings "design orientated work" into participatory research (ibid., 2012, p. 1). What sets more conventional Action Research apart from PD are the design activities, the creation of experiments, design artefact or design "things" (Ehn, 2008). Every design activity, process, or design artefact (or all of them) becomes a means to obtain and produce knowledge.

In this type of participatory Action Research strategy, knowledge production is embedded and braided into social action and creative activities (Greenwood & Levin, 1998; Archer, 1995). Action Research is carried out – similarly to PD – in partnership between researchers and people taking part in a collaborative research process. Routine Action

Research processes have reiterative cycles of diagnosing, planning, action taking, observation, reflection, and learning, as modelled by Susman and Evered (1978).

As mentioned in the introduction, Kurt Lewin coined the term Action Research in the 1940s and described it as "comparative research on the conditions and effects of various forms of social action and research leading to social action" (1946, p. 35). In tandem, Lewin suggested a cyclical and iterative research approach where data collection and action would be intertwined. Since its introduction, this cyclical model of Action Research has been re-developed into various versions of a cycle by many scholars in different disciplines, while the core elements have remained the same. In short, in the planning stage, the shared objectives are set, and the plan is developed accordingly. In the action stage, the plan is implemented while recognizing a need for flexibility to allow adjustments to be made, which is achieved through observing and assessing the process and its outcomes. In the reflection stage of the process, the researchers and participants "analyse, synthesize, interpret, explain and draw conclusions" (Kemmis & McTaggart [1988] 2013, p. 86) In addition, critical reflection is carried out through theory; this type of reflection is often directed at the research communities. Similarly to Action Research, in PD processes critical reflection plays a crucial part, especially through the lens of a *"reflective practitioner"* (Schön, 1984) and of *mutual learning*. When studying the design research process, Swann (2002) points out that the reflection of scholars "in action" and "on action" often leads to Action Research.

Evidently, Action Research has transformed the professional practices of PD scholars and practitioners. Yet a major shift in professional design practice happened when the role of the user in design

changed from being merely an informant providing implications and ideas to the process, to being a rightful member of the design endeavour. From the beginning, the PD approach has acknowledged people as experts in their own domain and has included and invited them to be part of the design processes (Schuler & Namioka 1993). As the movement advanced, scholars and practitioners sought means to enrich and enable greater participation, and eventually confirmed that 'everybody designs' and constantly redesigns in their everyday (Bannon & Ehn, 2012; Manzini, 2015). Increasingly, design researchers have argued, and reported through empirically grounded findings, that it is beneficial to braid together the users' domain with designers' domain and to develop "in-between spaces" and "in-between infrastructures" for collaboration (Muller, 2002,; Botero & Saad-Sulonen, 2010; Marttila & Botero, 2017). Furthermore, the technological advantages that have given people a new variety of possibilities to take part in design have expanded the design space and participants' abilities to explore it (Westerlund 2005; Botero et al., 2010).

Even if wider participation in design is easier than ever, in real life participants often have less power in the processes compared to professionals in the PD projects. It should be noted, however, that participants do not necessarily need to take part in all choices or design decisions for a project or its outcomes to be participatory (Bratteteig & Wagner, 2016a, 2016b. The field still continues, after 50 years, to include new theoretical perspectives and ever-developing methods on how to share responsibility and power, and how to open the design research process and practical decision-making. In this way, PD has also been a driver in joining theory with practical merits. To ensure that design sustains and supports participation rather than alienating and disempowering people, PD as a community

is increasingly developing novel strategies rather than solutions, and suggestions for others to use, change and develop. Healthy participation and community can be encouraged, not imposed.

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2.7 On Humans, Other-Than-Humans, and Technologies: Positions, Knowledges, and the Art of Paying Attention

Sisse Finken

When you engage in action-oriented research and practice Participatory Design (PD), you carry with you a concern – an ecological concern you could say – for both humans, other-than-humans, and technologies. That is, more specifically, when you practice PD you work closely with the ones who are going to use – or are going to be affected by – the new information technology imagined and/or designed. Along with this ecological concern, you step into empirical sites with an awareness that technology is not value neutral, but rather is designed with and from different preferences, politics, and values (Markussen, 1996; Haraway, 1988). Having such awareness also means that questions concerning who designs, for whom, how, and with whom, come to matter (Finken et al., 2018). This further means that elements such as involvement, influence, responsibility, and power are essential for such a design approach to ecologies of humans, other-than-humans, and technologies.

With this understanding of research and design, the door is open for considering research methods, which are brought along to design sites, to be value-laden just like technologies. You could go so far as to say that methods create knowledge (Finken, 2005). Meaning that methods include and exclude specific aspects in our observations,

our interviews, in our analyses, and in the very co-creating design work we engage in. Moreover, certain things are illuminated, while others fade into the background, precisely like the processes that go on when we observe and analyse the world through the theoretical lenses we use in research. Theory also matters. And it matters for action-oriented research. As can be read in chapter 2.6, PD has, beyond a strong focus on design methods, a solid basis in different theories, which assist in readings of and approaches to making that which goes on in co-creating design settings. In PD, therefore, you find examples where Wittgenstein (e.g. Ehn & Kyng, 1991), Phenomenology (e.g. Frauenberger et al., 2010), Feminist Scholars (e.g. Suchman, 2002; Sefyrin, 2010), and STS (e.g. Heitlinger et al., 2018) are shaping important work.

In this way, and in echoing Haraway (1988), there is no such thing as an innocent position, which is also to say that the bibliographies, positionalities, philosophies, and the (so-called) tools (research methods and methodologies) of researchers and/or designers create knowledges, which ultimately shape worlds of others, be it merely by descriptive accounts or by digital technologies. They shape realities.

In illustration, in chapter 3.1 you can find an instance where methods are not valuefree and do not create neutral grounds a priori. Further, to remain in this logic, and to include another example, you can say that the same goes for the methodologies we chose to use in our research, be it e.g. Case Studies (Stake, 2005), Ethnography (Madden, 2010) or Action Research (Avison et al., 1999). With such a view on methodologies, you can start to see them as research strategies that put our work and meetings with others on specific meaning-bearing paths. You can find an example on how a methodology sets the scene in a specific way in Boulus & Bjørn (2008), where they

present their initial reflections on Action Research. In the paper (presented at a Science and Technology conference) they consider their roles and work while taking part in an Action Research project working with different partners. One aspect they draw attention to, is the circumstance that the roles of both the researchers and the participants, and the very research focus, constantly develop and become when doing Action Research. They further highlight the need to be aware that, while immersed in doing Action Research, it might be difficult to step out of the project and reflect on one's own role and interventions.



Image 1: Empirical material from an ethnographic study of a smart-house for elderly residents in Oslo, Norway

A last note in this chapter on action-oriented research concerns the very art of paying attention to and being open to different opinions, conflicting viewpoints, and experiences. With this I give the word to Callén et al. (2009) who reflect on a research project concerning telecare and advocate for the use of Participatory Action Research in an effort to enhance participation. Among other important points, they refer to Spink (1999) and say that *"this task consists more of giving "ears" to those who cannot listen than of giving "voice" to those who cannot speak. Older people, disabled people, and all kinds of marginalized groups CAN speak"* (p. 121).

With this, I'll return to one of the introductory lines of this chapter and repeat that involvement, influence, responsibility, and power are essential aspects to consider when engaging action-oriented with ecologies of humans, more-than-humans, and technologies.

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2.8 Future Workshops: Turning the Affected into the Involved

Hanne Westh Nicolajsen

Future Workshops is a method developed by Robert Jungk and Norbert Müllert in the 1970s. They invented the method to involve multiple groups of stakeholders in the search for solutions to societal challenges.

The method alternates between phases of creativity and reasoning. The aim is to think big based on our dreams and aspirations in order to see beyond what exists. And then to develop solutions that suit us and create better futures. The participants develop plans, which support the implementation through specified activities and defined responsibilities.

The method builds on involvement of the affected, openness and democratic decision-making.

- The first phase involves *critique*, during which participants are invited to express criticisms, share negative experiences, and highlight limitations, thus alleviating any frustrations related to the topic
- In the subsequent phase, known as the *fantasy* phase, participants are encouraged to indulge in creative thinking, constructing utopian or wishful scenarios. This phase serves the purpose of broadening people's perspectives

- In the *reality* phase, participants formulate realistic scenarios based on the imaginative utopias developed earlier
- This is followed by the *implementation* phase, during which the requirements for actualizing these scenarios are discussed, encompassing considerations regarding the type and quantity of resources needed and placing responsibilities

The method received quite some traction in Scandinavia, probably because of its integrated democratic value. The method has been taken up both in the political system, in research, and in organizations as a democratic method to develop, innovate and make changes by including relevant stakeholders.

In research, Future Workshops may act as a method for Action Research. While it may support the creation of an intervention that includes a variety of stakeholders, it may simultaneously support insights into the perspectives of different stakeholders, relations and interactions within and across stakeholder groups, as well as concrete examples of challenges, aspirations, solutions, and conditions in play. All these insights are important data for understanding and developing an area of practice during research. Future Workshops have been applied as a research method across multiple fields such as Information Systems, Human-Computer Interaction and Participatory Design. A couple of examples of the method of Future Workshops applied in research settings are: involving users in software design (Kensing & Madsen, 1992), and investigating and supporting the provision of new library services (Nicolajsen et al., 2015). Or the case presented in this publication – the investigation and intervention within a Public-Private Research Project on the digitalization of a public home and health care.

The application of Future Workshops in research often twists the method. The recommended duration of the method is a number of days. This is however often changed into hours, and while the method was developed for a large cohort of heterogenous groups of multiple stakeholders, it is often used for fewer stakeholder groups or even a somewhat homogenous group of stakeholders. The method provides a shortcut to deep understanding of what is at play regarding the specific area undergoing change. In a short time, the participants and the researchers will gain a deep understanding of all the perspectives, stories and interpretations held by the participating group. A Future Workshop supports exchange of understanding and perspectives, and may also help in developing mutual understanding among the different groups, since it provides a vehicle for approaching a common theme in very concrete ways, which will expose different and better understandings of the positions of others. No matter if the group of participants is diverse, big or small, the method itself is strong in allowing participants to vent their frustrations.

When the frustrations are out, there is potentially a space to think beyond the extant and to dream. Our experience is that this may sometimes be difficult, and here facilitation is needed if some participants are too solution-oriented from the beginning. When the *fantasy* phase works, it allows for a deeper understanding of what is important to the participants. This may result in solutions that are based on deep values that lie beyond what seems immediately feasible. And the *reality* phase helps in testing assumptions about what is feasible, while the *implementation* phase is good when understanding the resources and responsibilities required, which again tests the feasibility and develops accountability.

A strong and weak point of the method is the mutual influence

between the participants, which may result in group think in good or bad ways. A way to mitigate this is by introducing time to reflect individually in each phase, and then taking turns in presenting the participants' individual thoughts. Thereafter, allowing for more rounds of input to get the ideas rolling, before the input is summed up, discussed, and prioritized. It is also our experience that an introduction by the facilitators before each phase is important to set the stage for what is required and allowed. This may also require a follow up to make sure that the participants do not discuss solutions too soon, and later in too fluffy a way. Another risk is power games. Even with rounds of turn-taking, there might be power games going on, which again require some facilitation, especially if there are participants who are in a hierarchical relationship. The inclusion of managerial staff might change the content, and the participation of managers should therefore be considered carefully. If managers participate together with subordinates this might need to be explicitly addressed and facilitated.

Overall, the strength of the method is the opportunity to vent frustrations and point out critical situations, the opportunity to dream without limits and then have a phase of realism that modifies the dream into something doable, something actionable that can fuel a change process. It opens a change project in a new way, bringing it close to the hearts and experiences of those involved. It is very creative but in a framed way, which again makes it actionable. Because of the personal involvement, the method is good at building engagement and thus preparing for action and change.



Image 1: The author's children's suitcase repurposed for Future Workshops.

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2.9 Action Research – and the Action Researcher, Inspired by Kurt Lewin

Hanne Westh Nicolajsen & Vasiliki Baka

Lewin's concept of Action Research posits a systematic approach involving the implementation of interventions followed by careful observation of ensuing changes. The subsequent reflective analysis serves as a catalyst for learning, with the acquired insights informing and guiding future interventions as well as theory building. The objectives of this method are thus twofold: firstly, to enhance existing practices, and secondly, to cultivate theoretical advancements.

The action researchers are involved in that they become part of the field of action. The action researcher can be an involved practitioner who also engages in research, or a researcher invited into fieldwork. Action Research as an approach comes as an alternative to positivistic research which claims that knowledge creation is necessarily objective and value-free. On the contrary, a key priority is how to create knowledge that is relevant and crucial to the well-being of individuals and groups, as well as to broader democratic social change. The desired change is realized through a cyclical process. The research process starts with diagnosing a need for change/ intervention, and is then followed by action revised through rounds of evaluations and modifications as crucial steps in the process (see figures 1 and 2 on next page).

Planning
Executing
Evaluation through Fact Finding
Modifying the 'Overall Plan'

Figure 1: The Action Research Steps, freely after Lewin (1946)

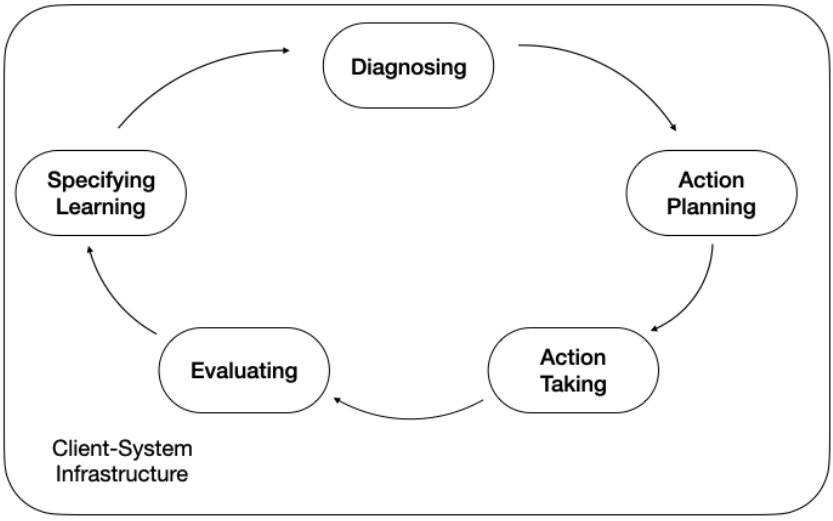


Figure 2: The Action Research Cycle, after Baskerville (1999, p.14)

This co-creative research process, which is very much shaped by both the researcher and the informants, is unique in that the roles are perhaps more difficult to differentiate than in any other research

approach. The different sectors and research fields add their own peculiarities and further define what the role of the action researcher becomes. The action researcher could be a full-time researcher who engages in a development process from planning to doing, and follows and analyses what happens. In fact, the observation and reflection processes are often the stages that the researchers are most knowledgeable about and engaged in. Researchers may thus provide expert knowledge to inform planning and doing at the very beginning, as well as following an intervention. In so doing, the researcher provides input for the planning and action phases, the subsequent interventions, and potentially also for the current theory on the topic of the Action Research cycle.

Apart from researchers who engage in Action Research, practitioners following e.g. a professional Master program often find themselves in the role of action researchers. This happens when they use their organization as a research case in the context of their education. In most cases they might design interventions based on new situations, or evaluations of existing processes having diagnosed problems and the need for change. This particular group mostly prioritizes planning and doing, while academic education strengthens their analytical and reflective skills.

As academics/lecturers, we many times also employ Action Research to different degrees. 'Action Research' in education was first introduced in 1954 by Stephen Corey. Mills, fifty years later (2003) defined it as *"any systematic inquiry conducted by teachers, administrators, counselors, or others with a vested interest in the teaching and learning process, for the purpose of gathering data about how their particular schools operate, how they teach, and how students learn"* (Nolen & Putten, 2007, p. 401). In the REFLACT

research group in particular, we have been involving students and practitioners actively in the learning process, like we did, for example, in the course 'Open Innovation'. In that context, the students were active co-creators of the processes, content and technologies used, while practitioners from the private and public sector co-defined problems of interest.

Outside of the educational context, all of us have been involved in various Action Research projects. In their research on democratic businesses, Nielsen, Nicolajsen and Baka brought entrepreneurs from cooperatives together to identify common topics and inspire knowledge sharing. Through this process of giving voice to the people affected, the action researcher becomes a facilitator and co-creator of problems and solutions. Baka (2017) employed Action Research with the aim to empower local people in Luanshya town, Zambia. Civil servants, residents and NGOs were actively involved in both defining and solving specific problems with the use of an interactive platform. The researcher thus acted as facilitator as well as technology expert in the process. In the cases presented in the following chapters, we share various other examples of different positions the action researcher can take. In Chapter 3.3, Dittrich et al. talk about the action researcher as the "neutral actor" in establishing ESG data reporting in the FinTech industry. In Chapter 3.1, Nicolajsen et al. talk about the researcher as an expert in the scaling and implementation of digital technologies in a public organization.

Irrespective of the context, doing Action Research is time consuming, and although the theory prescribes following multiple rounds of interventions, the process is by no means straightforward. Thinking in ongoing processes of interventions, as Action Research prescribes, is powerful, especially in projects where the end-results are difficult

to foresee, which is mostly the case when implementing new socio-technical practices that require ongoing work in organizations to make them work well. Hence, Action Research and the associated methodological principles of continuous learning, improving and adaptation, can help us understand digital technologies in organizations and society in more informed ways.

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3.1 A Future Workshop and Researchers as Experts: Action Research in a Public-Private Research Project

Hanne Westh Nicolajsen, Sisse Finken and Lars Rune Christensen

Virtual Care is a public-private research project, which has run over a five-year period, in which a municipality collaborated with a telecare solution provider to develop virtual home and health care through the use of a digital platform. The digital platform resembles Skype in that it is a form of video communication with a simple user interface to facilitate communication between the care-receiving citizen and the care-giving municipal worker.

The development of the platform was complex in that it both concerned development of the technology itself and the organizational practices. This meant that the development process emerged and took form as the practices and the platform evolved within the ecosystem of home and health care. In this way it is a socio-material ecosystem that expanded along the way.

From the perspective of the municipal home and health care area, the main concern was the co-creation of new practices by way of applying, adopting, reconfiguring and developing the new digital platform. The telecare service provider was concerned about developing the platform to fit the organizational needs of the municipal home and health care as the needs arose and

evolved. The telecare service provider wanted to understand these processes in order to accommodate other municipalities. In addition to redesigning the platform to local needs, they also looked for insights about the required maintenance and running of the system, as well as opportunities for establishing a support system for the implementation and running of the telecare platform.

Developing socio-material solutions, which entails both a digital platform, the practices, the maintenance and the support system, requires close collaboration between the partners. In this case it involved the telecare solution provider and the municipality's home and heath care management. Other involved user groups were those within home and health care in the municipality (care-givers, nurses, occupational therapists), the internal IT department, as well as the involved citizens, their relatives, and the broader ecosystem of general practitioners and hospitals (as the platform use developed and included them).

Researchers as Experts

In the Virtual Care project, the researchers had the role of socio-technical experts. While researching the initial phase of the project, the pilot, we, the researchers, provided analysis of the pilot implementation and took part in development meetings at the municipality. The municipality had previously experienced limited success with top-down management of scaling solutions from pilot implementations in one area to the rest of the organization. Thus, in the Virtual Care project, the municipality – inspired by the researchers – decided to apply a bottom-up approach to scaling by redoing the pilots in all regions. This process allowed for each of the regions to learn and grow in their own way, as they gained experiences with

respect to their different conditions. Experiences from each region were instead used as shared knowledge, which could inspire use of the platform in other parts of the organization.

Part of what made the bottom-up approach to scaling work, was the ongoing process of interactions between the central project leader and the local home and health care regions, supported by close contact with the tele solutions provider. These ongoing interactions allowed the innovations and the implementation to happen more easily, and with a higher implementation success rate than previously experienced. Eleven out of twelve regions implemented and re-innovated the practices to their individual modus operandi. The processes of re-innovation continued after the initial implementation, where employees were continuously engaged in the implementation and development of Virtual Care. Later in the process general practitioners and hospitals also got involved. Thus the project continued to expand as opportunities were revealed.

Future Workshop

At the beginning of the Virtual Care project, we, the researchers, conducted a case study in which we collected data from interviews and meetings, along with secondary material such as roll-out plans. However, throughout the Virtual Care project, inspired by Action Research, we sought to inform the implementation process by providing insights to make it run in a way that had greater potential to succeed.

Thus, along with the more traditional qualitative study and our aim of informing the implementation process as experts, we also decided to run a Future Workshop close to the end of the five-year project.

The aim of the workshop was to gain more nuanced insights into the collaboration between the municipality and the tele solution provider. We experienced a strong sense of loyalty between the partners in the project and found it difficult to make informants from the municipality and the tele solution provider talk about the challenges they faced in their collaboration. We thought that gathering everyone in the same room, looking more carefully at the collaboration project, would give the two organizations insights that could help them move forward, and could additionally give us insights about their different understandings of the project, the process, and the challenges, including what went well, and what was considered problematic.

For the workshop, we invited everyone involved at management level – both those involved in daily management of the project and those from the steering community from both organizations. The Future Workshop lasted two hours and included the phases: critique, fantasy and reality (the two latter phases of reality and implementation were collapsed into one (for an explanation of Future Workshops see chapter 2.6 in this booklet or Jungk & Müllert, 1987).

Running the Future Workshop

On the day of the workshop, the daily manager from the tele solution provider could not make it to the workshop. This gave an imbalance regarding participants. Three organizational levels of managers from the municipality were present, and two levels from the tele solution provider. At the beginning of each of the three phases of the workshop, the participants individually noted down their thoughts on post-it notes before they went up – one by one – to a large whiteboard to share their thoughts. When everyone had shared their viewpoints, the researchers facilitated a discussion, and

then asked the participants to group the input (on the post-it notes) in themes. The themes were then listed, grouped, and prioritized by the participants.

At the Future Workshop we used video recording (which we analysed later), took notes, observed, and facilitated the communication between the involved stakeholders. During the workshop it became clear how each participant thought about the different elements of the project's processes, mismatches in processes and mismatches in different terms used for the same thing, how they developed new ways of thinking, and how understandings gained during the project period informed their plans and actions. The outcome of the workshop brought along several surprising insights, both for us (the researchers) and the participants, as it became clearer how things worked and why. Consequently, the participants agreed to meet straight after the workshop to make some changes for their future collaboration.

In our facilitation of the Future Workshop, we added and practised strict turn-taking among the participants. Turn-taking was added to ensure that all participants got space and a voice. However, when we analysed the recording, we noticed how the three senior managers took over in subtle ways, which was invisible to us during the workshop. The way the situation played out limited the voices of those lower in the hierarchy. In the aftermath of analysis, we realized that the Future Workshop had become a negotiation between the partners at a strategic level. Our experience makes it apparent that power issues easily become part of the process and in particular when working with organizations and participants from different management levels. This in turn also affects the data generated and how research can come to inform practitioners.

With this, we stress the need to take note of ethical issues that could challenge the democratic element when running a Future Workshop, and/or, more broadly, when engaged in Action Research that should at best inform and further innovate processes like the one we have been following in the Virtual Care project.



Image 1: Future researchers from REFLACT

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3.2 A Case of Action Research with a Focus on Collective Learning about Organizational Challenges and Solutions for Democratically Owned Businesses

Lene Nielsen, Hanne Westh Nicolajsen and Vasiliki Baka

As researchers, we organized a two-day workshop and invited practitioners to share their organizational challenges. Researchers, together with practitioners, collectively created solutions to these challenges.

Taking Hayes' Action Research steps as a starting point (Hayes, 2014), we created the workshop focusing on 1) Establishing a Relationship with a Community Partner, 2) Research Questions and Problem Statements, 3) Action and Intervention, 4) Evaluation, 5) Disseminating Knowledge and Documenting Progress, 6) Leaving the Site (see chapter 2.4).

Looking back at the six steps, the process was as follows, which shows that it might not always be linear and not all steps might be included.

Before the intervention of bringing practitioners and researchers together, we worked on 2) Research Questions and Problem Statements. We had from previous research an initial knowledge of organizational challenges for democratic businesses, formed by

desk research and interviews with a larger group of practitioners, beyond the ones involved in the intervention.

The next step was to 1) Establish a Relationship with a Community Partner. In our case this was not just one community partner, but a mixed group of practitioners and researchers within the field, who would engage in the intervention.

This was followed by 3) Action and Intervention that consisted of both planning the intervention and asking the participating practitioners to formulate an organizational challenge before the workshop started.

Before the intervention (the two-day workshop) we had planned the following process:

- Collective input on the organizational challenges formulated by practitioners
- Frame the challenges and solutions in meaningful ways for practitioners
- Suggest possible ways to move forward
- In groups of three to four people, talk about the key takeaways from the day before
- Make overview of important themes
- Choose the theme you want to work with
- In the theme group make a poster with all issues related to the theme
- What action to take? When to take action? Who takes action?

Our approach had some deviations from Hayes' suggested process. In Establishing a Relationship with a Community Partner, we had

several partners from multiple research areas and from different democratically owned businesses working in different domains.

In the Action and Intervention, there was the possibility of mirroring between practitioners, learning from different challenges dependent on the situation of each company, and input from multiple research areas.

For Disseminating Knowledge and Documenting Progress, the knowledge was instant, and it created a deeper knowledge of the organizational challenges and the prerequisites for dealing with those challenges. Thus, the intervention was, in itself, a way to disseminate knowledge.

Evaluation was done on-site and included the key takeaways from the day as a form of evaluation.

For Leaving the Site, the practitioners gained a list of companies and researchers they could reach out to, while the researchers found people with similar interests. Thus, network and insights were gained.

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Image 1: Action and intervention – picture cards for participant introduction.

3.3 When Theory Becomes Action: The Data Commons Concept as Intervention

Yvonne Dittrich, Qiuyu Jiang, Cancan Wang, Laura Juncker-Jensen, Carsten Østerlund

Introduction

Action Research in cooperation with industrial actors often seeks to generate transformational change by introducing e.g. new methods or tools. However, in the case presented here, change started with proposing a different way of thinking, by introducing the Data Commons concept as an alternative theoretical frame for the collection of, reporting on, and benchmarking with data about Environmental, Social and Corporate Governance (ESG) practices. The research team invited actors in the FinTech domain to re-think ESG data from the bottom up. In a second step, we are now supporting the cluster organization Copenhagen FinTech as a core actor in the FinTech industry, in order to develop an approach to the collecting of data for a sector-specific benchmark for the Small and Medium Entities (SMEs) connected with it, which can be regarded as a stepping stone towards a commons-based governance of ESG data.

The research follows a flexible, reflective strategy (Bryant, 2002), rather than aiming at implementing predesigned intervention cycles. This allows the researchers to further develop both the concrete

intervention and the theoretical underpinning in dialogue with the industry partners and other stakeholders.

The project expanded over two years and in two distinct phases. After giving up our original plan of quickly getting to the design of the governance and tools for an ESG Data Commons, we focussed on understanding the wider context of ESG reporting. Here the concept of Data Commons both created traction and triggered innovative thinking in the FinTech community. It became a conceptual probe to investigate diverse perspectives on ESG data reporting. In the second phase, the Copenhagen FinTech cluster became the subject and coordinating organization for the subsequent data collection and benchmark initiative. Here, the research team accompanied the initiative rather than driving the change.

Below we introduce the theoretical basis we developed as part of the research and then shortly present both the phases and the result, with a conclusion about the power of theoretical concepts as intervention.

ESG Data Commons

ESG reporting, the disclosure of non-financial information about business impact in the areas of environmental, social and corporate governance for investors, customers, and wider stakeholders (United Nations Global Compact, n.d.), is an integral part of the green transition of trade and industry. Traditional ESG reporting is regulated though national and international legislation (Wang et al., 2023). However, the legally required reporting mainly targets bigger companies and is perceived as both too generic and too cumbersome for SMEs. The resulting poor ESG data quality often impedes effective ESG reporting.

Our project aims at addressing these challenges by applying a Data Commons perspective, which regards data as a jointly governed and used good. We adopted a practice-view, which allows us to approach Data Commons as an ongoing process, where a nexus of doings and discourses unfolds over time and space (Schatzki,1996). We conceive it as an ongoing collective becoming that produces and reproduces an ecosystem of data and metrics among a network of participants (Wang et al., 2022). The ESG Data Commons refers to a collaborative platform where stakeholders within the FinTech industry share, co-create, and manage ESG data. Our practice-based approach allowed us to explore various aspects of an ESG Data Commons:

- Change Process: ESG Data Commons is viewed as an ongoing process, evolving over time as practices adapt to contingencies. It continually transforms, defined as much by what it's not as by what it is
- Teleoaffective Structures: Teleoaffective structures guide stakeholders' practices, driven by goals, beliefs, emotions, and motivations. Different stakeholders may have diverse goals and motivations when collaborating around data
- Sociomaterial Dynamics: The approach integrates social and material aspects, highlighting the role of objects, including infrastructural, activity, epistemic, and boundary objects. These objects evolve and play various roles throughout the development of a Data Commons

Phase 1: Exploring challenges and possibilities of ESG reporting using Data Commons as a conceptual probe.

The ambition of Phase 1 was to start designing and building an ESG Data Commons for the FinTech domain. We therefore involved representatives of Denmark's Export & Investment Fund (formerly Vækstfonden) as a user of ESG reports and benchmarks, and a sustainability data technology provider called Tekudo.

Phase 1. 1: Planned and emerging process

Our research design was initially conceived as an Action and Design Research approach, encompassing a series of well-defined stages as e.g. proposed by Baskerville (1999): Preliminary research would (1) develop the basis for action planning, and (2) in the form of structured design workshops. The workshops and a possible implementation of the results would comprise the implementation phase (3), which would be followed by further observation (4). This cycle would provide a well-documented base for reflection and publication (5).

However, things turned out differently: When starting the preliminary research, we realized that an ESG Data Commons for FinTech needed to involve a larger and more diverse community than expected, leading to adjustments in our research design.

We continued our investigation by iteratively identifying key stakeholders and interviewing them regarding their perspective on the current state of ESG reporting, and the challenges and potentials they saw. In the interviews, we used the Data Commons concept as a probe to test and refine our ideas (Brandt et al., 2012). The conceptual probe can also take a material form such as a stakeholder map, where we ask our informant to add or modify stakeholders and their relationships based on our initial mapping (see Figure 1). The research team then followed up on these changes with probing questions aimed at understanding why and how specific stakeholders are part of the ESG Data Commons. This process resulted in Copenhagen FinTech's decision to develop an ESG benchmark through a bottom-up data collection in the second year of the project.

In this process the concept of Data Commons served as a conceptual

probe that changed how ESG data are discussed in the Copenhagen FinTech cluster which focusses on SMEs, supporting their growth and participation in the FinTech domain.

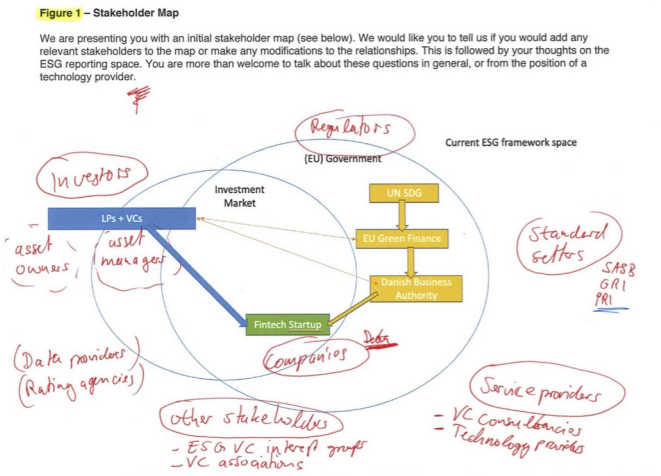


Figure 1: Stakeholder map as a probing device for ESG Data Commons.

Phase 1. 2: The ESG Data Commons as a conceptual probe

As an innovative and alternative approach to data management, Data Commons created a space for reflection in the discussions. Our interviewees started to critically reflect on their own practice and considerations. The interviews started to address stakeholder-specific requirements and challenges instead of discussing advantages and disadvantages of current approaches.

In turn, we needed to extend our understanding of ESG Data Commons: ESG Data Commons need to involve not only FinTech startups and financial service providers, but a wider range of stakeholders, including venture capital firms (VCs), who need to

justify investments to their customers; government entities, who develop regulations for ESG reporting; and cluster organisations, who equip the SMEs to be successful in this context. Further, ESG reporting can draw on other data reporting practices and data reported in different contexts. In other words, the concept of Data Commons facilitated an iterative process of 'problematizing' and 'problem-solving', that helped gain pivotal insights and uncover hidden aspects of ESG reporting.

In this process, the concept of Data Commons became an intervention in its own right, facilitating innovative thinking and bringing together a very diverse set of stakeholders to explore a more meaningful approach to ESG reporting.

It also became visible that it was important that the researchers were perceived as impartial: Several different actors told us that as researchers we were well-positioned to create impactful ESG metrics and practices. We were perceived as 'neutral actors' that represent neither the state nor the market, and were expected to bridge the different interests of various industry actors. In other words, we researchers opened up a design space for ESG reporting for the FinTech domain (Dittrich et al., 2014)

Phase 1. 3: Concrete result of Phase 1

Together with our partners, we gained a better understanding of the gaps between ESG frameworks adopted by investors, partners, and FinTech startups, as well as EU regulations. We have also gathered an overview of the services provided, daily operations, and data capabilities of FinTech startups. Beyond the theoretical results, three concrete results served as steppingstones for the ESG baseline study that became the focus of the second year.

First, to help SMEs make sense of ESG reporting, we developed an 'ESG Compass' (See Figure 2) containing an overview of the current FinTech ESG landscape. The ESG Compass breaks down generalized and non-specified ESG reporting requirements into different principles pertinent to the specific company, i.e. its motivations or incentives (know-why), its stakeholders' foci (know-who), its most important metrics (know-what) and the connection of these to its business operations (know-how) (Wang et al. 2023).

Second, we documented our own learning in the report 'ESG in Fintech: An Overview' (Wang et al., 2023) that provides a rich



Figure 2: Excerpt from the ESG Compass that provides an overview of the current FinTech landscape. For full version, see: https://issuu.com/reflact/docs/esg_in_fintech_compass

source of material, links to relevant legislation, and documents the perspectives of the relevant stakeholders. The report is both an introduction to ESG reporting for researchers and could be used a resource for companies.

Third, we used the knowledge about existing ESG reporting frameworks and FinTech startups to build an 'ESG Checklist' (see Figure 3) – a survey to collect ESG metrics adjusted to the character of FinTech companies and the services they provide. It consists of 50 questions covering the environment and the social and corporate governance of a FinTech company. The adaptation does not only concern what ESG is about, but also how ESG can be related to the daily operations of a FinTech startup. For instance, the questions are categorized under the headings of 'team', 'financial product', and 'governance'.

In summary, the process of building an ESG Data Commons is an ongoing collective effort (Dittrich et al., 2002; Teli et al., 2017) that aligns multiple communities, goals, and aspirations while reconciling differing perspectives and imaginaries.

Phase 2: Baseline Study as a Probing Device

Phase 1 helped us to identify the specific challenges of FinTech startups regarding ESG reporting: a mismatch between the general data requirements and daily operations of FinTech startups; a lack of understanding of how to find and report relevant data; volatility of the operations of the startups; and mismatches when comparing companies across industries.

To address these challenges, the demand for an ESG benchmark

	Question Type	Data Type	Category	Operationalised Question
Basic Info				
Time	Open-ended	Number	N/A	When was the company established?
Size	Open-ended	Number	N/A	How many employees, including full-time, part-time and interns, are currently employed at your company?
Board	Open-ended	Number	N/A	How many board members are there currently at your company ?
Fiancial Service Provided	Open-ended	Text	N/A	What kind of financial service do you provide in the company?
Investment Stage	Open-ended/Multiple Choice	Text	N/A	Which investment stage is your company currently in?
Respondent	Open-ended	Text	N/A	What is your current role at the company?
Team				
Gender Diversity - Founders	Open-ended	Percentage	Social	What is the percentage of non-male founders?
Gender Diversity - Workforce	Open-ended	Percentage	Social	What is the percentage of non-male employees in the workforce?
Gender Diversity - Board	Open-ended	Percentage	Social	What is the percentage of non-male board members?
Gender Diversity - Workforce Pay Ratio	Open-ended	Percentage	Social	What is the workforce pay ratio?
National Diversity	Open-ended	Percentage	Social	What is the percentage of international employees in the workforce?
Racial/Ethnic Diversity	Open-ended	Percentage	Social	What is the percentage of racial/ethnic group representation for the workforce?
Age Diversity	Open-ended	Percentage	Social	What is the age diversity ratio in the workforce?
Diversity and Inclusion Goals	Dichotomous	Yes/No	Social	Does the company have diversity and inclusion goals?
Diversity and Inclusion Learning	Dichotomous	Yes/No	Social	Does the company participate in diversity and inclusion learning to understand what works and what does not?

Figure 3: An in-progress version of an 'ESG Checklist'.

that evaluates the current ESG status of companies and projects frequently appears in our interviews with legislators and venture investors, and in pension funds' initiatives, as a necessary step for identifying a common goal and a shared definition of 'good ESG' in the startup and FinTech communities. The diversity of available ESG frameworks and stakeholders in the FinTech ecosystem meant that it was difficult for actors with different interests, such as investors, regulators, and startups at various stages, to agree on a common goal for 'good ESG' for the cluster. In addition, ambitious stakeholders who are keen to push for ESG reporting in FinTech startups find themselves debating questions such as "which metrics are the most authoritative", or "in what way ESG would make sense to small startups that are not legally required to do ESG reporting". These debates are often based on various assumptions about the startups' desires, resources and capabilities, and uncertain future trends in the field of ESG, often without resulting in any fruitful conclusions.

The baseline study that assesses the current state of ESG in FinTech startups thus becomes relevant for us as an effective 'probing

device' to understand these assumptions in practice among the stakeholders in a FinTech cluster. That is also why we jumped on the research opportunity, when the CEO of Copenhagen FinTech welcomed an ESG benchmark study across its member startups.

Phase 2.1: Research design and choices

During Phase 2, we invited a venture investor and an ESG data provider to join us as new project partners. Whereas in Phase 1 we relied on Copenhagen FinTech to suggest partners to us, in Phase 2 we screened and selected companies that had the expertise we needed to carry out the benchmark study, such as, expertise in emission calculation of IT services, and specialized experiences with ESG reporting in startups. In return, the new partners were incentivized to support the benchmark study via the hope for creating a push for tech startups to identify common ESG goals that would also benefit their own operations. Meanwhile, Copenhagen FinTech emerged as an important collaborator in the project due to their mediating role in the FinTech startup community. During the pilot study and the planning of the benchmark study, together with Copenhagen FinTech, we jointly recruited companies through their communication channels, and discussed the value and the processes of the baseline study.

In February 2023, we distributed the ESG survey to four FinTech companies, who expressed interest in incorporating ESG practices into their operations. The aim here was to test the design of the questionnaire. The founders of the companies filled in the questionnaire physically in a room where the research team was also present. The research team explained the questions and interviewed the respondents when specific questions were not clear. This pilot study resulted in initial ESG reports for individual companies, as well

as data visualizations that illustrated the performance of an individual startup on the various ESG parameters relative to the cluster averages for easy comparison. Furthermore, it also highlighted the overlap between the FinTech-specific ESG categories and existing reporting categories for financial licensing, to help the company understand the connection between ESG reporting and company operations.

At the time of writing, we have reported the results of the pilot study (with a select sample of cluster companies) to Copenhagen FinTech and adjusted the questionnaire according to the early learnings, in preparation for rolling it out across the whole industry as the first ever Danish FinTech sector ESG benchmark study. Currently we are coordinating how to carry out the benchmark study, hopefully involving all companies in the Copenhagen FinTech cluster.

Phase 2.2: Probing through conducting the ESG baseline study

Moving from theoretical concepts to concrete data collection and analysis not only requires time for the technical implementation, but it also comes with several new questions: What values can a variable take and what is an average value? How should the data be visualized in an actionable way? A collaborative research process involving industry insiders helped us explore the concept of Data Commons and address the doubts it raised during Phase 1. This gave rise to new understandings of how 'probing' is conducted as part of Action Research, and notably what it means in a moving field like ESG and startups. Our ongoing reflections revolve around three conditions that made the probing possible: 1) researchers and organizational participants as boundary spanners (who), 2) the importance of documentation as a boundary object (what), and 3) the timing of probing as embedded responses (how):

Researchers and organizational participants as boundary spanners (who).

Boundaries and power negotiation between knowledge communities became visible as we inquired into the practices of the FinTech startup communities through the concept of Data Commons. This required researchers and project participants to invest in boundary spanning work that "build(s) common ground between separated parties" (Levina & Vast, 2013.) Our role as a 'neutral actor' (see above) opened doors for us to access different communities. A broad range of participants were eager to share their practices and influence the novel field of ESG. To establish a sustainable Data Commons that has a chance to continue existing after the project, it was imperative to collaborate with an existing boundary spanner in the FinTech entrepreneurship community, Copenhagen FinTech. Copenhagen FinTech also shared our vision for sustainability in the FinTech domain, and was already well-positioned in the community to push for change. They were able to help us translate our ambition across the field and had weight in pushing for new sustainability-oriented business practices.

The importance of documentation as a boundary object (what).

Both finance and startup communities are often characterized by high velocity, volatility, and a lack of time. Artefacts that can be extended over time thus play a critical role when communicating and collaborating with actors from these communities (Østerlund, 2008). In our experience, documents such as one-pagers that summarize and communicate research ideas are important objects of exchange between the researchers and the FinTech startups. One-pagers are also valued in professional communities as symbols

of professionalism and conciseness, which create a shared format for researchers and organizational participants who are not familiar with each other's work practices. More importantly, it affords a 'mutual probing' opportunity for both researchers and participants to reflect on and challenge each other's assumptions. For example, when communicating our baseline design, we mapped out key research design areas such as 'samples and populations', where we suggested our demarcation of the population of companies and sampling methods. Meanwhile, we left questions such as "What is your ambition for the use of the baseline study in the future?" as ways of facilitating decision-making and co-creation with Copenhagen FinTech. The one-pagers were thus used not only as a mutually familiar format for communicating progress and planning, but also as a tool for sharing ownership among the partners by indicating open questions to be decided. This has proven to be a strong facilitator of co-creation and knowledge sharing.

The timing of probing as embedded responses (how).

Timing is an important condition for probing, as the responses from the participants rely on their everyday work practices as well as the broader context. However, it is a condition that needs to be carefully observed and attended to rather than controlled. At an early stage in our study, we realized that our research design was often conditioned by the work events and schedules of startups, investors, and Copenhagen FinTech. We could not carry out a study during certain periods because significant events for the FinTech communities were taking place such as 'Tech BBQ' or 'Nordic Fintech Week'. Nonetheless, our study did coincide with an important social movement focusing on ESG that interested all involved stakeholders. Despite the practical limits of the research project, new regulations

and initiatives acted as important impetuses for participants to engage in our study.

Conclusion

What struck us the most when reflecting on the research described above, was the *effect* that the *concept* of Data Commons had: It changed the dynamics in the interviews from evaluating existing approaches to discussing alternative ways of looking at ESG reporting, and the needs and challenges the different actors faced. In the transition from Phase 1 to Phase 2, both the interview results and the summary of challenges resulted in Copenhagen FinTech's willingness to implement a baseline study and, as part of that, develop technical infrastructures for a potential future Data Commons.

In other words, the introduction of a theoretical concept became a powerful intervention, bringing heterogeneous actors together to address a common challenge. It resulted in one of the core actors, Copenhagen FinTech, taking their action further, implementing a baseline study and at the same time creating the technical prerequisite to implement an ESG Data Commons.

This might not be traditional and structured Action Research 'by the book'. Instead, it required constant reflection on our own roles as researchers in relation to the ongoing development, a careful and continuous deliberation of next steps in collaboration with the involved communities, and an accountable documentation of our actions and those of others. In our case, the Action Research was never meant to be traditional and structured if we were to achieve the intended impacts, and the methodology that resulted from the study in our view serves to move the field of Action Research forward.

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Reflections on Change-Making in Ethnographic Research on the Green Transition

Vivian Wei Guo

Introduction

Humanity faces a triple planetary crisis of climate change, environmental pollution, and biodiversity loss. International governments, business organizations, and civil society groups have started to advocate for the importance of the 'green transition' – a shift towards environmental and climate-friendly economic development (UNEP, 2022). But how should researchers, especially those who are trained for researching long-term, immersive lived experiences, approach such topics concerning planetary crises?

I argue that an action-oriented ethnographic research design can provide ways forward, that is, an ethnographic approach that can both account for and enable transformative change. I would advocate that it can contribute to solutions to the planetary crises in two ways: first, by providing a critical account of the promises and changes introduced by societal and technology actors, to ensure their actions are relevant and accountable for dealing with these crises; second, by doing so, our research can also identify and provide opportunities for change in the ways organizations and technologies are operated.

This chapter presents my reflections on this question from my early-stage PhD research, where I investigate how non-governmental organizations (NGOs) in China take part in sustainability governance, using data about the Environmental, Social and Corporate Governance (ESG) performance of businesses. I will start by introducing the research background: ESG data and the green transition in China. I will then present my research design, study case, and current stage of the research. I conclude this chapter with reflections on my aspirations for, and the challenges I encountered when conducting action-oriented ethnography.

Research Background: China, ESG Data and the Green Transition

Since 2020, ESG reporting has gradually become interchangeable with sustainability in the corporate parlance. It refers to various reporting frameworks and standards for assessing a company's performance in the areas of Environmental Protection, Societal Impact, and Corporate Governance (OECD, 2020). These frameworks and standards are created by various organizations including governments, NGOs, investors, business groups, and think tanks, in order to regulate, supervise, monitor, and evaluate companies with the aim of improving their sustainability performance. Some internationally acknowledged ESG reporting frameworks and standards are The Global Reporting Initiative (GRI) Framework, The Task Force on Climate-Related Financial Disclosures (TCFD) Framework, Sustainability Accounting Standards Board (SASB) Standards and European Sustainability Reporting Standards (ESRS). Information about a company's ESG performance, that is, ESG data, especially serves as a critical compass for companies, governments, investors, and other stakeholders when evaluating the sustainability and ethical impact of a business, influencing investment decisions and corporate practices.

As a key player in global governance for green and sustainable development, China has a significant influence in shaping the future landscape of responsible investment and business practices. Internationally, it has issued a policy for promoting green overseas business and investment under the Belt and Road Initiative. Domestically, it has announced the 'dual carbon' policy, aiming to reach peak emissions by 2030 and carbon neutrality by 2060. Overall, the Chinese government has ambitions to shift its economic development towards a greener track.

The ESG measurement framework is envisioned to play a critical role in allowing for the implementation, monitoring and evaluation of China's governmental policies and claims. The Chinese government has thus put a great effort into improving ESG data management. Its journey towards a Chinese ESG system has involved gradually establishing regulatory measures and guidelines to promote sustainable business practices. For example, the China Securities Regulatory Commission has issued ESG disclosure requirements for listed companies, and guidelines for financial institutions to integrate ESG factors into their risk assessment and decision-making processes (Shen et al., 2023). It has also been actively seeking to localize international ESG standards to address local conditions and development goals (ICMA, 2023).

Research on ESG Data and Environmental Organization in China: An Action-Oriented Ethnography

I adopt an ethnographic approach to my research design (Stewart, 1998), where I draw on the methods of participatory observation and interviews (Davies, 1999). I am particularly interested in exploring local ESG data-related initiatives, and local expectations,

perspectives, and reactions towards ESG and sustainability-oriented changes in practice. Empirically, I follow a Shanghai-based environmental organization, QY, that promotes climate change mitigation and environmental protection through monitoring, developing, and governing corporate ESG data.

By the time of writing, I had done preliminary fieldwork. During my fieldwork, engagement in change-making had become a very relevant point of reflection – that is, finding myself contributing to the change that is being promoted by the actions I study. Nowadays, it is difficult not to engage in change-making when studying a community such as environmental organizations that constantly seek for ways to deal with planetary crises such as climate change. This focus on change-making adds an interesting dimension to my considerations about the research design, especially taking into account what is typically understood by ethnography. An ethnographic methodology, often "seeks primarily to understand (not change) the conditions of the community being studied" and sees the world from the point of view of those studied. Despite this, ethnographic research has been seen to be either occupied with or implicated in change and change initiatives from the start (Barab et al., 2004, p. 254-255; van Willigen, 2002). Nonetheless, in this sense, ethnography stands in stark contrast to change-oriented methodologies such as Action Research, that explicitly aim to induce change, increasingly with explicit democratic, participative, and interdisciplinary values (Jones et al., 2023).

There is no doubt that there is a sense of urgency in researching our planetary crises. The question is, is there a way to incorporate an Action Research mindset, allowing change-making to enter our understanding of ethnography?

In Egmose, Hauggaard-Nielsen and Jacobsen's article 'Action Research in the Plural Crisis of the Living: Understanding, Envisioning, Practicing, Organising Eco-social Transformation', the authors argue that, in dealing with eco-social challenges, we need a rethinking of the relations between humans and nature and how they are carried out into practice (Egmose et al., 2022). They especially highlight the importance of "learning from and with human practice and experience" (p. 677), as well as the process of co-creation of knowing with people (ibid). In this line of thinking, Action Research methodology can be particularly relevant for understanding the role of ESG data in relation to individual, organizational and institutional change-making, because it can help identify and induce avenues for change regarding ESG data practice. This could be a way to steer societies towards a more sustainable and equitable direction, while providing individuals, organizations, and relevant societal stakeholders with critical learnings on how sustainability-oriented changes take place in businesses.

Participatory Design researchers have argued extensively in favour of ethnography taking on a role in their field as an action-oriented methodology, which can better inform design and "bring about change that is defined by the interests of workers, the requirements of their work, and the jointly negotiated path to change" (Blomberg & Karasti, 2012 p. 11-12). More specifically, Blomberg and Karasti argue that combing ethnography and Participatory Design, and iterating between them with reflexivity, can be an effective way for the ethnographic researcher and designer to learn about the organization, and generate a realistic vision of the future uses of technology for changing concrete practices (ibid.).

Drawing on these schools of thought, I propose that our role as

researchers in studying the planetary crises is double-layered. Apart from providing understandings through our engagement with our informants, we are also advocates for promoting, and facilitators of sustainability-oriented changes. In the context of my PhD project, the ethnographic fieldwork is not just a way to collect data, but also a mutual learning process in which myself and my informants understand local desires, practices, and requirements of ESG data. That is, while putting forth my understanding and observation of the matter, I help raise my informants' awareness of their own assumptions and reflection on their practices regarding ESG data, so that they can rethink and adapt these to better address local sustainability problems.

Application of Action-Oriented Ethnography in My Preliminary Fieldwork

The way this approach played out specifically in my preliminary fieldwork from October 2023 to February 2024, was that during this period when I was based in Copenhagen, I had online conversations with the managing director and founder of QY, where we discussed collaboration forms and made an initial collaboration agreement. During our communication, he explicitly mentioned his interest in EU ESG regulations (e.g. Carbon Border Adjustment Mechanism) and data standards (e.g. European Sustainability Reporting Standards) and asked for my perspectives on the implications for China. Following our communication, I started to follow the development of EU ESG regulations and data standards, particularly through collecting data from news media and scientific papers, as well as participating in relevant conferences and workshops in Europe. For example, I attended the European Commission's EFRAG annual conference 2023 'European Corporate Reporting – Two Pillars for

Success', and several online public and private EFRAG events on European Sustainability Reporting Standards (ESRS). My relationship with the director of QY evolved due to my participation in these EFRAG activities on ESRS in particular. And my work with interpreting ESRS became a key topic of our research collaboration. It was due to this interaction that I became particularly interested in the concept of standards and standardization in the global-local context. I started to reflect on how the EU requirements regarding ESG data may have contributed to shaping climate and data governance in China, and how the EU ESG standard and its standardization approach may have implications for actors beyond the EU, including China. I furthermore engaged these reflections in my currently ongoing fieldwork in Shanghai from February to June 2024.

During the fieldwork in Shanghai, I have taken part in QY's daily organizational work and observed everyday practices, including how they conduct ESG data analytics, organize and participate in relevant ESG events, and provide ESG and sustainability training and data services to external stakeholders. This was combined with interviews with QY's staff and its main collaborators and relevant stakeholders. In doing so, my goal was to gain insights into these data professionals' understandings, operations and visions regarding climate change and eco-economic decoupling, as well as the relationship between climate change and IT. That is, how they articulate climate change as a problem in demand of a response enabled by ESG data, and how they practically apply IT with the climate in mind.

Overall, this action-oriented ethnographic research process is strategically designed but also evolves somewhat organically as the interactions between researchers and informants evolve (Lin et al., 2007). I move fluidly from the role of a researcher to the role

of a change facilitator, which, in turn, strengthens my informants' interest in continuing to be engaged as research participants and, for example, help me gain access to the local community. However, I am also aware of the possible methodological and ethical challenges I might face, such as potential conflicts of interest and the shrinking distance between me as a researcher and my subjects, which I will discuss more in the next section.

Challenges and Aspirations in Conducting an Action-Oriented Ethnography

Moving forward with my action-oriented ethnographic approach during the next phase of my fieldwork in China (February–June 2024), I hope to contribute to a constructive, reflective, and balanced narrative that describes and discusses the problems and conflicts that emerge around ESG data at both the individual and organizational levels. In this way, my research aims to contribute to a better understanding of local priorities, challenges and power structures that impact ESG discourses and data practices in the green transition.

In practice, I am deepening my participation in the local translation and standardization of ESG data. I am participating in local ESG data-related initiatives and experimentation, and I am observing local professional practices regarding ESG data. Drawing inspiration from Action Research, I will use the method of Future Creating Workshops to understand eco-social relations and create social learning spaces involving local ESG data practitioners and other relevant stakeholders (Egmose et al., 2022). Future Creating Workshops aim to encourage critical and visionary thinking on everyday life experiences, as a way of creating new relevant knowledge and nurturing an environment

for social change and transformation (ibid.). Using Future Creating Workshops, I hope to provide a "free space" (Egmose, 2015) to facilitate participants in analysing their own existing eco-social relations that are grounded in their everyday lived experiences. It is also the aim that they become able to identify local problems and activate social imagination, by creating visions and narratives that can direct future action regarding ESG data, and enable learning from doing practical experimentation (Egmose et al., 2022).

Doing action-oriented ethnography can give rise to various challenges. During my preliminary fieldwork, I have tried to initiate a relationship with QY and gain access to the field and social networks by proactively reaching out to the director of QY and other relevant stakeholders. During this process, I have struggled with my identity, sometimes oscillating between a researcher on the one hand and an ESG professional who is actively working with QY on the other, and sometimes being both at the same time. Such an experience can be confusing when choosing how to present myself in certain settings and networks. Furthermore, I have found myself constantly evaluating the level of participation that is suitable to research agendas where one usually needs a certain distance from the research informants, and I have also had to consider the influence that the evolving relationship with the director of QY has had on me.

Critical reflective observation is crucial to ethnographers for understanding "the nature of their participation and its suitability to the particular research circumstances, and the relationship between researcher and subjects." (Davies, 1999 p.74). Participation in people's daily lives facilitates observation of certain behaviours and arrangements and can build a good foundation for open and meaningful conversations with informants (ibid.). As an ethnographic

researcher, I consider myself as alternating between four different roles in the field depending on different situations and conditions, namely: complete observer; observer-as-participant; participant-as-observer; and complete participant (Gold, 1958; Davis, 1999). I started my fieldwork by applying the conventional strategy of impression management to secure access to the field sites and field relations for collecting data (Hammersley & Atkinson, 1995). Later, however, I started to experience the complexities of my own identities, roles and relationships that have characterized my fieldwork. I also became aware that the fieldwork engaged me in identity work, which can be both productive and problematic (Davies, 1999; Coffey, 1999). Moving forward, I will pay more attention to the evolving relations between the field settings, research participants and the private self, as well as to how fieldwork shapes and constructs identities, relations, an emotional self, and a physical self (Coffey, 1999). I plan to use such reflections as a productive device to peek into the identity formation of my informants as I observe their practices and partake in their daily activities and organizational life.

Overall, through action-oriented ethnography, I hope to provide inspiration, lessons, and experiences for instigating sustainability-oriented changes in business practice. Furthermore, I wish to provide understandings of how individuals and organizations can lead more sustainable practices through digital technologies. This is also a call for more of my fellow ethnographers to engage in and share their experiences with conducting action-oriented ethnography research that investigates green transitions and sustainable transformations in practice. This, I hope, will allow for developing a combined ethnographic and change-making practice for studying and promoting the green transition.

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